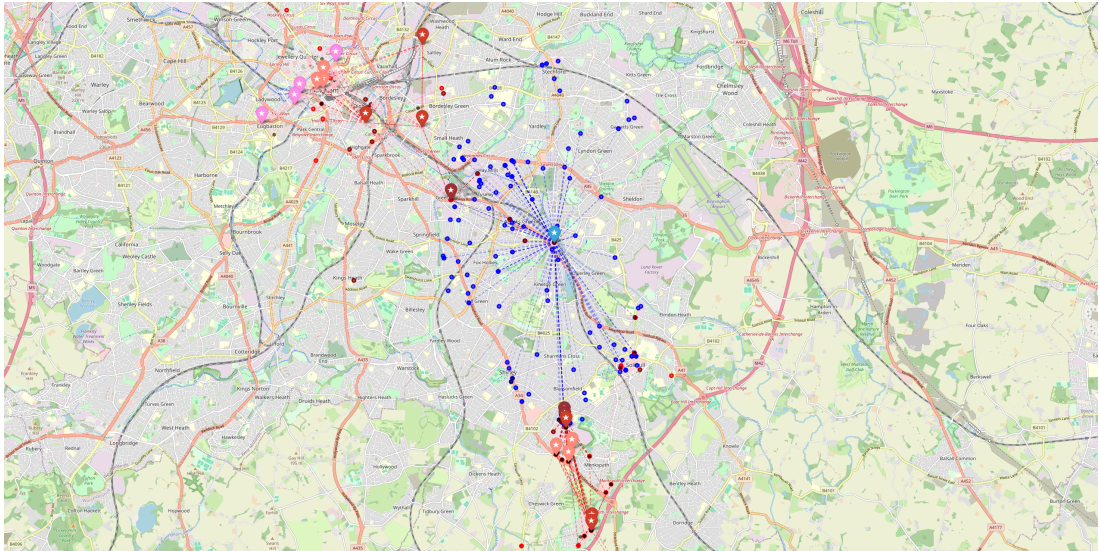


Estimating spatial firm productivity spillovers using UK microdata



Firms are spatially interconnected. But are they interconnected when it comes to productivity?

Aloysius Lip ^{1 2}

Dissertation Supervisor: Professor Lars Nesheim ^{1 3}

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University College London (UCL)

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¹ University College London (UCL)

² HM Treasury (HMT)

³ Centre for Microdata Methods and Practice (CeMMAP), Institute for Fiscal Studies (IFS),
Economic Statistics Centre of Excellence (ESCoE)

DECLARATION

I, Aloysius Lip, hereby declare that the work presented in this dissertation is my own original work. Where information has been derived from other sources, I confirm that this has been clearly and fully identified and acknowledged. No part of this dissertation contains material previously submitted to the examiners of this or any other university, or any material previously submitted for any other assessment.

I declare the use of GenAI tools in an assistive role (UCL Category 2) for this submission. The model Gemini 3.10 Pro (Google DeepMind, 2026) was used for the limited roles of: supporting the particular process of reviewing and refactoring code (data analysis), and as a support tutor (economic model and empirical approach). The model GitHub Copilot (Microsoft, 2026) was used for the limited roles of: supporting the particular process of reviewing and refactoring code.

Name: Aloysius Lip

Date: 2nd September, 2026

CLASSIFICATION

This piece of research is primarily:

- ☒ an empirical/econometric study
- ☐ the examination of a theoretical problem
- ☐ a critical analysis of a policy issue
- ☐ an analytical survey of empirical and/or theoretical literature

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WORD COUNT

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ABSTRACT

This paper estimates the productivity spillover occurring between colocated firms in the United Kingdom. The productivity spillover, a spatial externality, underpins well-established economic phenomena around the development of urban clusters in the New Economic Geography literature, and is measured and heavily discussed in the evidence on the economic effects of urban density measured in the spatial econometrics literature, and underpinning regional growth policy.

I develop *prima facie* evidence for the spillover by developing a 3-factor production model of Total Factor Productivity (TFP), capital, and labour, with TFP in turn decomposed to spillovers of peer TFP, capital, and labour. I apply recently-developed econometric techniques for valid identification in peer effect group and network settings to comprehensive UK firm data from Fame. The key to valid identification rests in internal instruments from spatial *exogenous* effects in a static model, and from deep lags in a dynamic model: this overcomes Manski (1993)’s ‘Reflection Problem’ and the endogeneity of peer effects. It requires big data calculations of firm interactions, which I use modern data techniques to compute efficiently, preserving the advantages of Fame’s rich balance sheet microdata for both its panel structure and its resulting high-significant estimates.

Across my instrumented specifications, I evaluate a positive spatial spillover from the labour inputs of a peer firm of (0.4? Calculation is complicated), onto any given target firm’s productivity. I evaluate a milder positive spatial spillover from a larger accumulation of capital inputs from the peer firm. The implied spillover from any peer firm’s TFP to a target firm is negative, (implying that firms compete for local demand(?) and in any contemporaneous period, one neighbour’s success detracts from their peers). Contrasted with models assuming strict exogeneity across in all peer input decisions, which suggest only positive spillovers from peer factors to TFP, this suggests the need for caution in devising regional economic policy. Spillover effects are moderate and conditional, require peer firms to increase their factor inputs specifically. A multiplier effect via a continuous chain of firm-to-firm induced positive productivity shocks is less empirically supported.

This analysis establishes a baseline framework on which more sophisticated, and necessary, extensions to investigate the phenomenon of agglomeration can be built. These include the implementation of network and industry fixed effects and patterns of measurement to examine the heterogenous strength of the productivity spillover, informing how and where agglomerating effects may be strongest.

1. INTRODUCTION

The strength of the spatial productivity spillover is crucial to policy and economic discussions on growth, production, and the formation of urban clusters, but it is notoriously tricky to estimate⁴. The purpose of this paper is to demonstrate that modern advances in rich microdata compilation, computing, and econometric theory and techniques mean that progress is possible in this decades-long empirical problem. Indeed, evidence is derived with a stylised, classical model only using features available from firm data, which provides a baseline for future more sophisticated models of households, urban development, and endogenous selection. This analysis offers four key contributions that are substantively novel:

1. The development of data science techniques to calculate spatial transformations on rich microdata to fulfil exogenous regression specifications
2. The application of higher-order spatial lags as an internal instrument deploying a peer effects model in a firm productivity setting
3. The application of higher-order temporal lags as an internal instrument in a dynamic panel peer effects model, contrasting model performance with these two instruments
4. Empirical estimates for a firm-to-firm spatial productivity spillover. This is evidence that can underlie patterns of firm colocation discussed in the more sophisticated urban development literature, and provides a framework for future extensions to investigate heterogeneous effects.

This analysis acknowledges a large body of work on agglomeration, rooted historically in Marshall (1920), which elucidate the importance of urban density and development on firm productivity. In section 2, I give an overview of the substantive contributions from urban economics and New Economic Geography (NEG), which explain agglomeration driver channels and offer sophisticated approaches to modelling to household and firm dynamic location choices. No attempt is made to recreate this whole body of work. Instead, this analysis abstracts away from the usual components of urban density and dynamic decision models. Despite being a spatial analysis, location is barely employed as a feature in the analysis of productivity except to identify each firm's spatial peers. From this mere spatial proximity, this analysis looks for empirical evidence isolating the pure productivity spillover from peers to a target firm. It therefore may provide empirical support for the centrifugal, pro-urbanisation force underlying the more sophisticated analyses of New Economic Geography and regional development.

⁴ Qualitative discussions in Krugman (1998), Fujita and Krugman (2004)

Section 2 also gives a brief overview of the econometric literature on peer effects model, from Manski (1993)’s ‘Reflection Problem’ to finding valid instruments in a more complex graph and network setting in Bramoullé et al. (2009). This analysis leans heavily on that econometric theory to develop a model of peer effects for the productivity spillover between firms.

In sections 3 and 4, I discuss the handling of data from Fame, a database of publicly-available, balance-sheet data from UK companies, unified and designed to be comprehensive for research. The cleaned Fame dataset provides more than 100,000 firms’ data across many years. This is extremely useful for deriving precise econometric estimates, but a significant data science challenge to compute and manage the memory for vectors of millions of observations and their transformations. In these sections, I highlight my use of data science techniques to calculate the spatial weighting matrix $\mathbf{W}$ and the iterated evaluation of its higher-order spatial lags of $\mathbf{W}$, such as $\mathbf{W}^2, \mathbf{W}^k, (\mathbb{I} - \mathbf{W})$, on over 1 million observations. This offers a reusable method to calculate valid instruments and conduct transformations that are helpful for eliminating endogeneity. I provide a detailed guide to reproducing data methods in Appendix A and all code for this analysis is available publicly⁵.

In section 5, I give the econometric analysis of a static model with spatial peer effects, conceptualisable as ‘spatial lags’ of the classical regressors in a 3-factor production function. This implements econometric theory from Bramoullé et al. (2009) for the identification of higher-order spatial lags as valid instruments to the *endogenous* regressor. I estimate models under both a ‘groups’ peer structure, which argues that the strength of a productivity spillover is the average of nearby neighbours, and a ‘networks’ model, which applies an individualised weight for each target-peer firm connection proportionate to the reciprocal of the physical distance between them.

In section 6, I expand the use of internal instruments to higher-order temporal lags. This builds on modern ‘System GMM’ methods from Blundell and Bond (1998). These include an autoregressor and exploit the time variation of the panel structure to instrument for both the *endogenous* peer effect components and all other regressors, which have their assumptions relaxed to weak endogeneity. These models offer limited evidence of a dynamic autoregressor coefficient, or significant coefficients for lagged peer effect variables, validating the use of static models which restrict effects to the contemporaneous period.

Throughout this analysis I prioritise extensions, additional groups, lags, and instruments, that expose the econometric properties of the model, rather than being designed to make them more policy-relevant. I propose that the baseline of

⁵ <https://github.com/alloysiuslip/ucl-diss-ECON0079>

establishing the method of investigation, with credible contributions on the data science, theory, and econometric method of this issue are sufficient for the scope of this paper. Policy-relevant extensions are discussed in Appendix 6.2. I leave it for further analysis to build on the framework I establish this paper and seek to investigate true, policy-meaningful questions that I discuss in this introduction.

1.1. (PRODUCTIVITY IN THE UNITED KINGDOM)

1.2. (SPATIAL PRODUCTIVITY TO POLITICAL ECONOMY)

A revival of interest in regional, pro-second city policy has also taken place in the policies of the 2024 and 2026 governments in the United Kingdom. Regional economic policy declined strongly in the UK under the 1980s Thatcher government (...)

This consensus was maintained throughout the 1990s-2000s New Labour period, where rapid economic growth known as ‘The Great Moderation’ satiated impetus for change. The consensus was broken in 2017, with the seeds of a regional-industrial strategy in 2017, emerging under the Conservative May government. Under the background of a UK-EU Referendum result (“Brexit”) which emphasised the sentiment of disaffection among ‘left-behind’ places across the UK which did not see the benefits of national aggregates of economic growth primarily concentrated in the UK’s capital, London. The national government launched ‘Local Industrial Strategies’ which sought to identify action to "boost productivity, earning power, and competitiveness" on a regional basis⁶. However, fewer of these strategies were fully developed and implemented by 2021.

In 2026, the newly appointed Burnham government made the pledge for “good growth in every postcode”⁷ with immediate speculation and interpretation from economic commentators⁸ on what this might mean for the direction of regional and urban policy in the United Kingdom in the absence of further details.

⁶ Saywah Mahmood and Georgina Hutton and Lewis Pickett (2025)

⁷ GOV.UK (2026)

⁸ Matthew Gill (2026), Muthu De Silva (2026)

2. LITERATURE

2.1. (AGGLOMERATION AND MARSHALL)

It has been common to analyses of economic growth since Marshall (1920) to assume and qualitatively discuss some effect on firm-to-firm spillover of productivity. A productivity spillover underpins well-established economic phenomena such as the colocation of firms in urban clusters described in New Economic Geography literature such as Krugman (1991). It has been heavily analysed through industry and spatial density approaches in spatial and urban econometric literature such as Glaeser et al. (1992).

My investigation is rooted in Marshall (1920)'s taxonomy, which identifies three channels to productivity spillovers: (1) input-output costs, (2) labour pooling, and (3) knowledge spillovers. There is no reason to think these are exhaustive. We might theorise that there may be competitive effects and cultural factors. Krugman (1991) also labels all these channels as 'centripetal' to an urban core; there are likewise 'centrifugal' factors that push firms away from the urban core, such as destructive (zero-sum) competition, congestion, and saturation costs. A further extension could develop models on these further channels and empirically test them. Furthermore, these productivity spillovers are incredibly salient to thinking about production structures.

In this investigation, I concentrate on (3) knowledge spillovers, tackling a difficult identification problem highlighted in Glaeser (2001). This is more relevant to the knowledge-based services economy that dominates the UK. Input-output costs and labour-pooling matter more for industrial manufacturing firms, especially the latter which was devised in response to the phenomenon emergence of industry towns.

Most agglomeration literature bundles all three channels together. Glaeser (2010) and Krugman (2010) identify how vital identifying knowledge spillovers are. That need to estimate them well only grows with the modern, knowledge economy of the UK. (Goldin, 2023) speculates that agglomeration effects have "increasing strength" attributable to the way in which "knowledge workers in particular become more productive when located near to one another. Information may be more available than ever, but the evidence suggests that it still tends to flow largely through in-person networks".

2.2. (NEW ECONOMIC GEOGRAPHY)

However, Krugman (1991) prefers not to assume this effect in their focus on the dynamic movement of firms. Instead, heterogenous cluster policies, such as initial external differences in the number of workers or firms are frequently assumed,

with ‘forward and backward linkages’ driving colocation endogenously from there. This paper helpfully tests the presence of spillovers, providing another centrifugal force directly incentivising firms to colocate.

2.3. (MODERN APPROACHES)

Specific approaches to investigating knowledge spillovers via innovation, instrumenting via patent publications comes from US evidence, including (Jaffe, 1993), (Berkes and Ruben, 2021) and more recently Bloom et al. (2013).

Knowledge spillovers have been a well-evidenced factor of managerial literature within the firm contributing to productivity gains. A field-starting MIT study from Allen (1977) observed an exponentially declining curve depicting the frequency of communication between two individuals against their physical distance in an office.

2.4. PEER EFFECTS ECONOMETRICS

The Manski (1993) model is a common baseline in literature on peer effects. It provides helpful structure to the study of joint behaviour of individuals who are members of a group or network, and can be applied to any examination of units with peers such as individuals, not merely firms.

$$y_i = c + \underbrace{\alpha_g}_{\text{correlated effect}} + \beta x_{i,g} + \underbrace{\delta \bar{y}_{-i,g}}_{\text{endogenous effect}} + \underbrace{\xi \bar{x}_{-i,g}}_{\text{exogenous effect}} + \varepsilon_i$$

The model firstly decomposes spillovers to a direct behavioural, or *endogenous* effect, $y_j \rightarrow y_i$, where the outcome of interest among peers affects that of the target individual. Robust applications of the Manski model demonstrate credible approaches to measuring the *endogenous effect*. Damm and Dustmann (2014) analyses movers to new neighbourhoods among individuals through the ‘Danish Dispersal Policy’ of refugee arrivals (1986-1998). Chetty et al. (2016) measures a similar effect among high-poverty children through the US 1990s ‘Moving to Opportunity’ scheme. These studies exploit quasi-randomness in the location assignment of individuals and peers for robust identification. As discussed in section ??, the dataset used for this analysis unfortunately offers no such granularity of features to identify movers, let alone evaluating the exogeneity of their colocation. The use of firm fixed effects and reasonable assumptions of location stickiness are employed to compensate for this risk of location endogeneity.

Additionally, the Manski (1993) model requires contextual or *exogenous* effects, of which the variation of non-outcome of interest variables among peers also directly affects that of the target individual, $x_j \rightarrow y_i$. This brings sophistication to the

nature of peer spillovers. A simplified model only allowing endogenous effects through the variable of interest, $y_j \rightarrow y_i$ is nested within this structural model, but not identified due to the ‘Reflection Problem’ and as described in this analysis in section 4.1. In this analysis’ firm productivity model, the *exogenous effect* allows for an additional level of input factor capital and labour to induce higher target TFP, better management practices or more efficient combination of factor inputs, perhaps from observing the mere high level of these factors in their peers, without the peers themselves necessarily experiencing a productivity increase. It is additionally a common specification requirement for every individual to have an own-effect of exogenous characteristics, $x_i \rightarrow y_i$, which corresponds to the standard production function model in this analysis of $k_{i,t}, l_{i,t} \rightarrow y_{i,t}$.

$$\begin{aligned}
y_{i,g} &= c + \alpha_g + \beta x_{i,g} + \delta \bar{y}_{-i,g} + \xi \bar{x}_{-i,g} + \varepsilon_i \\
\bar{y}_g &= c + \alpha_g + \beta \bar{x}_g + \delta \bar{y}_g + \xi \bar{x}_g + \bar{\varepsilon}_g \\
(1 - \delta) \bar{y}_g &= c + \alpha_g + (\beta + \xi) \bar{x}_g + \bar{\varepsilon}_g \\
\bar{y}_g &= \frac{c + \alpha_g}{1 - \delta} + \frac{\beta + \xi}{1 - \delta} \bar{x}_g + \frac{\bar{\varepsilon}_g}{1 - \delta} = \bar{y}_{-i,g} \quad (\text{large } g \text{ assumption}) \\
y_{i,g} &= \frac{c + \alpha_g}{1 - \delta} + \beta x_i + \frac{\xi + \delta \beta}{1 - \delta} \bar{x}_{-i} + \frac{\delta}{1 - \delta} \bar{\varepsilon}_g + \varepsilon_i \quad (\text{substituting}) \\
y_{i,g} &= \alpha_g + \pi_0 + \pi_1 x_i + \pi_2 \bar{x}_{-i} + v_g
\end{aligned}$$

The focus of Manski (1993)’s linear-in-means (LMM) contribution is to highlight the ‘Reflection Problem’, an econometric trap. The derivation give above shows that the initial structural equation cannot be estimated directly with OLS due to the endogeneity of $\bar{y}_{-i,g} : \mathbb{E}[\bar{y}_{-i,g} \varepsilon_i] \neq 0$. Transforming the equation to reduced form and omitting the endogenous regressor leads to underidentification: 3 parameter estimates yielded for 4 structural parameters. Causal variation in $y_{i,g}$ is unidentifiable as it is exactly collinear with all the other regressors. This underidentification is a famous one, hindering analysis of peer dynamics including via spatial interaction.

A key solution to this is the deployment of instruments, $z_{-i} \propto \bar{y}_{-i} \cap z_{-i} \not\propto y_i$. An application employing credibly exogenous instruments for a peer effects measure in an individual crime setting is given in Dustmann et al. (2024). This study uses the sex-identity at birth of newborns as a valid instrument for the criminal behaviour of their babyfather gang-leaders, who serve as an influential peer on the target members of their gangs respectively. The equivalent valid instrument in this analysis which is correlated with the peer effect but has no direct effect on the target outcome of interest is an additional spatial lag of *exogenous* regressors in the static model. This means the capital and labour of a firm k who is a

‘peers of peers’: a peer of firm j , but not of firm i , the last of which is the target firm of interest. In this analysis’ dynamic model, deep temporal lags $y_{-i,t-2} \propto y_{-i,t} \cap y_{-i,t-2} \not\propto y_{i,t}$ are also as a valid instrument under an assumption of weak exogeneity of predetermined regressors.

A modern paper, Bramoullé et al. (2009), explores different identification techniques and their rules for validity for group and network structures. These structures are more complex than the Manski (1993) model, which by using a variable of means $\bar{y}_{-i,g}$, implicitly assumes simple, ‘leave-one-in’ group averages as the peer *endogenous* regressor. Moffitt and Caspi (2001) makes a modification to drop the ‘large g assumption’ and instead exclude the target observation $y_{i,g}$ from calculation of the group mean y_g , which provides a helpful conceptual move from estimating a ‘neighbourhood effect’, for which the target is still a contributor, towards more of a collection of influential peers. Nevertheless, this is still insufficient to offer identification.

The paper subsequently explores many complex interaction structures summarised by the weighting matrix $\mathbf{W}$, the n -dimensional interactions matrix between firms. I discuss and illustrate structures of $\mathbf{W}$ in detail in section 3.4. The structures of $\mathbf{W}$ are key to identification. The contributions from this paper are, in short, that complex $\mathbf{W}$ matrices which vary group sizes (s_g) and network diameters (s_d) are key to discovering valid spatial lag instruments. Fortunately, these structures resemble realistic interactions common to empirical data, and are applicable to this analysis’ pattern and layout of firm interactions. We should note in application that this strategy continues to assume exogenous assignment of spatial assignments and so target-peer interactions.

The identification rules from Bramoullé et al. (2009) are summarised in Table I. Critically, once certain necessary conditions for the group and network structures are met, an additional spatial lag on the *exogenous* regressors, $\mathbf{W}^2x$, tends to be a valid instrument. This analysis uses Table I’s approaches to instruments to strip out the endogeneity of a spatial peer effects model.

TABLE I
Spatial econometric identification rules for weighting matrix $\mathbf{W}$

Structure	Required condition	Identified	Instrument
<i>Group structure</i>			
Proposition 2	$\xi_0 - \delta_0\beta_1 \neq 0$ and $\zeta_0 - \delta_0\beta_2 \neq 0$		
Linear-in-Means ^a	Target is own peer	×	—
Leave-one-out ^b	Target excluded from peer mean	×	—
	Same group size		
Varying group sizes ^c	Target excluded from peer mean ≥ 2 group sizes	✓	$\mathbf{W}^2x$
<i>Network structure</i>			
Proposition 1	$\mathbb{I}, \mathbf{W}, \mathbf{W}^2$ linearly independent $\beta\theta_0 + \xi_0 \neq 0$	✓	$\mathbf{W}^2x$
Intransitive triad	$\exists i, j, k : \mathbf{W}_{ij} \neq 0, \mathbf{W}_{jk} \neq 0, \mathbf{W}_{ik} = 0$	✓	$\mathbf{W}^2x$
Deep network	$s_d \geq 2$ network diameter	✓	$\mathbf{W}^2x$
<i>Adding correlated effects</i>			
Proposition 4	$\mathbb{I}, \mathbf{W}, \mathbf{W}^2, \mathbf{W}^3$ linearly independent $\xi_0 - \delta_0\beta_1 \neq 0$ and $\zeta_0 - \delta_0\beta_2 \neq 0$	✓	$(\mathbb{I} - \mathbf{W})\mathbf{W}^2x$
Group structure	$s_g \geq 3$ group sizes	✓	$(\mathbb{I} - \mathbf{V})\mathbf{W}^2x$
Network structure	$s_d \geq 3$ network diameter	✓	$(\mathbb{I} - \mathbf{V})\mathbf{W}^2x$

Adapted from Bramoullé et al., 2009 with notation revised and abridgments to improve relevance to this analysis. (a) Manski, 1993, in this model $\mathbf{W}^2 = \mathbf{W}$ for linear dependence. (b) Moffitt and Caspi, 2001, in this model, $\mathbf{W}^2 = \frac{1}{s-1}\mathbb{I} + \frac{s-2}{s-1}\mathbf{W}$ for linear dependence. (c) Lee, 2007. Structures under a **Proposition** header must follow the **Proposition**'s required conditions. x denotes an *exogenous* regressor, in this firm analysis this means $k_{i,t}$ or $l_{i,t}$. It is comparable to the vector of regressors $\mathbf{x}_{i,t}$ that I use to denote the vector of both of them.

3. DATA ANALYSIS

3.1. FAME

I use data from Fame for this investigation. Fame is a comprehensive database of UK firms with balance sheet variables, which can be summed to useful aggregates for economic analysis: GVA, capital, and labour inputs. It is provided by Moody’s Analytics, an international credit agency, and collects its data from UK firms via survey⁹. Its international counterpart, Orbis, offers a similar service for international counterparts. This analysis could therefore be extended to offer insights comparing the spillover effects of UK firms to their international counterparts. For this investigation, given the background discussion laid out, the UK dataset is sufficiently comprehensive and policy-relevant.

Over 100 GB of raw data were accessed and downloaded from Fame in February 2025 by Professor Lars Nesheim for Irastorza-Fadrique et al. (2026) and reused for this project. Raw data were ingested to *DuckDB*, an open-source, MIT-licensed relational (SQL) database architecture¹⁰ optimised for high-compute variable manipulation. This is crucial for the applications of $(n \times n)$ matrix transformations to a variable n -length column vector, as I detail heavily for calculating peer weights and high-order transformations.

Every firm is tagged with raw, textual location properties. I construct a hierarchy of favourability to parse these, in order: *latitude/longitude*, *full address*, *address* (split over multiple fields), and *postcode* only. I additionally favour data listed for the firm’s *primary trading address* over its *registered office address* (headquarters). When only postcode data was available, I matched postcodes to the ONS’ postcode lookup database¹¹, yielding a decimal longitude-latitude value for the centre of its area. I compiled these heterogeneous features to 7 location identifier columns. The first set of these indicate membership of 3 spatial location groups: travel-to-work area (*ttwa*) using the 2011 definitions¹², the first-half of a UK postcode (*pc4*), or a full UK postcode (*pc8*). The latter 2 are point distance decimals for longitude + latitude. *longitude_decimal*, *latitude_decimal*, and 2 supporting columns indicating the data quality.

The resulting time-invariant table has exactly 152,379 firms with unique Companies House *registered_number* identifiers.

⁹ Moody’s Analytics (2025)

¹⁰ Mark Raasveldt and Hannes Muehleisen (2026)

¹¹ Office for National Statistics (2025)

¹² Travel-to-work area definitions from 2011 are still considered the mostly reliable for current analytical research, due to the pollution in travel and working patterns resulting from the COVID-19 pandemic during the 2021 Census. Travel-to-work-area area coverages are mostly time-invariant; changes between 2001 and 2011 were reasonably limited, see: Office for National Statistics (2015).

3.2. DERIVING PANEL AGGREGATES

All financial data are deflated to real terms, using 2024 as the base year. I use CPIH¹³ to deflate *renumeration_employees* and other income-related aggregates capturing household spending patterns. I use the GDP deflator¹⁴ for all other balance sheet variables. Financial data are in thousands of GBP (£'000s); as I estimate log relationships exclusively, this has no effect on regression coefficients. I compute these variables for the exactly 1,128,490 panel observations. Firms have a mean of 7 years of observations per firm. Further detail on the cleaning procedure and derivation of variables is available in Appendix A.1.

Labour input is the number of employees reported on the FAME balance sheet per firm. This dataset does not include firm-level hours worked per employee that would offer granular productivity (GVA per hour worked) measures. An estimation in this setting would require imputing the measure from industry aggregates, which would likely confound the true firm-level signal.

Capital is Fame’s *fixed_total* variable, which represents the sum of tangibles, intangibles, and other investments within the balance sheet, namely an extremely comprehensive measure of the firm’s fixed capital. The average capital-output ratio is 6x within FAME when filtering for positive values Table II, which is moderately higher than macroeconomic aggregates typically suggest likely due to accounting standard differences.

TABLE II
Capital-output ratios in FAME corpus of firms

Variable	Description	Capital-output ratio
<i>tangibles</i>		3.53
<i>fixed_total</i>	<i>tangibles</i> + <i>intangibles</i> + <i>inv._other</i>	6.00
<i>total_assets</i>	<i>fixed_total</i> + <i>current_assets</i>	11.56

GVA is calculated as the sum of *ebitda* and *renumeration_employees*. *renumeration_employees* is the exact sum of *wages*, *social_security_costs*, and *pensions_costs* in the balance sheet. *ebitda* is in turn *turnover* minus intermediate inputs and is given in the Fame balance sheet for the vast majority of panel observations in our dataset. It is also possible to reconstruct *ebitda* via other balance sheet metrics, but these sub-fields are actually available for fewer observations than *ebitda*.

¹³ Office for National Statistics (2026a), CPIH is the ONS’ preferred price index for household consumption Office for National Statistics (2026b)

¹⁴ Office for National Statistics (2026c)

To help guide the choice of aggregate measures and understand the observation structure, I conduct a benchmark 2-factor production estimation in panel structure to help validate the choice of aggregates against known measurements.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + u_{i,t} \quad (1)$$

This specification does not yield consistent estimates when estimated with OLS, according to the structural model I give in section 4.1. $u_{i,t}$ is a non-structural error-residual correlated with $k_{i,t}$ and $l_{i,t}$. Nevertheless, it is a helpful exploration tool to observe if derived coefficients match macroeconomic estimates

TABLE III
Benchmark 2-factor TFP estimation

	(1.) <i>gva</i>	(2.) <i>gva</i>	(3.) <i>gva</i>	(4.) <i>gva</i>	(5.) <i>gva</i>	(6.) <i>gva</i>
constant	2.995 (0.036)	3.036 (0.031)	1.970 (0.010)	1.973 (0.010)	4.751 (0.018)	5.268 (0.070)
<i>employees</i>	0.617 (0.005)	0.614 (0.004)	0.561 (0.002)	0.561 (0.002)	0.720 (0.004)	0.628 (0.015)
<i>total_assets</i>	0.286 (0.005)	0.284 (0.004)	0.421 (0.002)	0.421 (0.002)		
<i>fixed_total</i>					0.065 (0.002)	
<i>tangibles</i>						0.090 (0.006)
<i>intangibles</i>						0.025 (0.002)
<i>investments_other</i>						0.007 (0.001)
<i>i</i> fixed effects	✓	✓			✓	✓
<i>t</i> fixed effects	✓		✓		✓	✓
Observations	1,045,164	1,045,164	1,045,164	1,045,164	1,006,904	152,444
R ² (excl. F.E.)	0.3313	0.3453	0.8094	0.8096	0.2830	0.2699
R ² (Overall)	0.7912	0.7900	0.8096	0.8096	0.6751	0.7403

Coefficients derived in Table III give a labour income share, $\hat{\beta}_2$, exactly in line with ONS analysis of UK industry-level data over the same time period, particularly *corporate gross labour share* which is constructed from a similar sample to Fame¹⁵.

I explore specifications which omit fixed effects. In Model (2.), the capital share of income jumps on the omission of firm fixed effects as all within-firm variation that we would decompose as $\alpha_i \in \varepsilon_{i,t}$ is instead attributed to capital. This gives us some insight that in this panel setting, the measure of capital as a stock variable is reasonably consistent within a firm over time.

¹⁵ Office for National Statistics (2024)

When varying regressor measures, surprisingly, running the regression with *fixed_total* as the capital measure instead of *total_assets* almost entirely eliminates the capital share of income, or explanatory power of capital on variation in GVA. Similar results are shown for a further decomposition of *fixed_total* to its asset components. I leave it for future analysis why this dataset yields this phenomenon and for the rest of the analysis, I use *total_assets* as the primary capital measure.

3.3. FINANCIAL AND LOCATION DATA LIMITATIONS

Following data cleaning procedures discussed in Irastorza-Fadrique et al. (2026), I removed duplicate entries, limited observations to those with more than 10 employees, and observations with the required balance sheet properties, *renumeration_employees* and *ebitda* required to calculate firm GVA.

Fame has rich microdata observations with granular detail about each firm in the UK, but it is not designed for spatial analysis without advanced data science techniques, improving even beyond the sophistication demonstrated here. There is some miscoding in Fame’s location data. Despite it being my favoured *amalgamated address* selection choice, Fame’s native *longitude* and *latitude* data contained some errors, such as recording either one of those 0.000 coordinates, swapping them, or placing them outside the UK. I employed several techniques to search for outliers. One instances is calculating the centre-point of all firms sharing a postcode and then filtering for firms further than a threshold: I evaluated that 1.8% of all firms in the cleaned sample are more than 5km away from their postcode’s centre-point. Due to the time and data processing constraints of this analysis, I dropped these firms and observations.

Though I recorded the quality of location data, I undertake no further analysis using it as its incorporation. I discuss further uses in section 6.2 to control and investigate the effects of miscoded location data and heterogenous treatment effects. These requires complex data science and Machine Learning approaches, which I leave for future analysis.

I amalgamate multisite firms to their primary trading address, as an assumption. In the dataset prepared for Irastorza-Fadrique et al. (2026), each firm is only tagged with their last-known registered address and primary trading address. Fame includes further trading addresses for multisite firms available for download through their data portal, however the only site-level datapoint provided is *number_of_employees* per trading address, with sparse and inconsistent coverage¹⁶. This analysis simply takes financial aggregates listed for each firm and assumed it to be located at the listed primary trading address. There are clear flaws to

¹⁶ Moody’s Analytics (2025), platform accessed August 2026.

this method: large retail firms, such as consumer retailer ‘Tesco’, have several hundred listed trading addresses on Fame, and generate much less than the full amount of GVA at their listed primary trading location in ‘Welwyn Garden City’. Nevertheless, difficult algorithm allocation decisions, or the pursuit of more sophisticated data sources are required to address this issue, as discussed in section 6.1. The amalgamation assumption provides ease of computation while contributing to attenuation bias of the estimated spillover parameters due to imperfect peer assignment.

Additionally, the location data that is available is time-invariant, a characteristic of the *fixed* table, rather than updated on a yearly basis to reflect firm-movers. This is a reflection of using a comprehensive dataset of financial datapoints, that was not optimised for spatial analysis; instead Fame provides an in-built feature suggesting peers via balance sheet and industry similarity, which I ignore for this analysis. One interpretation of this analysis’ use of these characteristics is that each firm is over the entire 20-year sample with no movers. This seems very unlikely: subsection 2.2 ‘(New Economic Geography)’ tells us to expect the endogenous location of firm movement and colocation to optimise for taking advantage of firm spillovers.

Using the location characteristics as I do, as a noisy signal of the firm’s true location, has 2 econometric consequences. Firstly, firm fixed effects in all spatial specifications become extremely important. For all movers within the sample period, an endogenous firm location decision can only be modelled as correlated with the spillover regressors of interest. This justifies my use of fixed effects over random effects throughout this analysis. Firm fixed effects are time invariant, and can additionally help to eliminate some of the endogeneity induced by movers who undertook their move just before the sample period, 2006-, to maximise their spillover effect.

Secondly, I acknowledge some attenuation to the productivity spillover estimate. Incorrect location data due to any of the above-discussed issues causes the assignment of spatial peers to any target to be miscalculated, and vice versa. Peer firms who were close to the target firm during the firm-year observation who subsequently moved away within-sample will have an extreme distance recorded, and so be excluded from tagged address groups and truncated distance calculations. Peer firms who are recorded as close to the target firm as they subsequently moved to the location. This reduces down the strength of the spillover coefficient θ_0 estimated.

3.4. LOCATION GROUP CLASSIFICATION

Throughout this analysis, I use terminology of a productivity spillover occurring from a group or average of *peer* firms onto a *target* firm. The influence of the vector of all firms j on the vector of all firms i can be summarised in an $(n \times n)$ interactions matrix, $\mathbf{W}$. A weight $w_{ij} \in \mathbf{W} \neq 0$ implies a firm-to-firm interaction, namely some use of the peer firm's attributes in the regression to estimate the target firm's TFP.

This interactions matrix is symmetric: any firm with a peer weight calculated by the distance between the two firms naturally is a peer in return with an identical distance and weight. Extensions to this analysis could devise asymmetric relations, for example, with bigger firms having more influence. The interactions relationship is irreflexive - no target can be a peer of itself. This means the diagonal elements of $w_{ii} \in \mathbf{W} = 0$. I summarise the properties for $\mathbf{W}$ in Table IV, with further discussion for $\mathbf{W}_g$ and $\mathbf{W}_d$ to follow.

TABLE IV
Weighting matrices i, j -relation properties

Matrix type	Symmetric	Reflexive	Transitive	Overlapping	Blocks
Group, $\mathbf{W}_g$	✓	X	✓	X	g
Network, $\mathbf{W}_d$	✓	X	X	✓	$ttwa$

The definition of a group within $\mathbf{W}_g$ is intuitive: peers interact in a fixed limit of nearby firms, where each peer within the group is treated as homogenous, offering an equal weighted value, here $1/(s_g - 1)$. The group matrix $\mathbf{W}_g$ is transitive: if firms A and B share a spatial group, $ttwa$, $pc4$, or $pc8$, then a firm C that is a peer of firm B must also be of firm A. Apart from its irreflexivity, $\mathbf{W}_g$ is block diagonal. I construct initial weights by peers which share the spatial group $pc8$, so each firm has peers of all the firms in its postcode, each weighted with the reciprocal of the number of firms in the postcode (leaving-one-out).

Formal rules for construction of the group matrix are summarised below¹⁷. I illustrate an example $\mathbf{W}_{g*}$, a (7×7) set of interactions between firms which have group classifiers of size $s_1 = 4$ and $s_2 = 3$ only in Figure I.

¹⁷ Adapted from Bramoullé et al. (2009)

$$w_{ij} \in \mathbf{W}_{g^*} = \begin{cases} 0, & \text{if } i = j \\ \frac{1}{s_g - 1}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbf{\Gamma}_{s_1} \in \mathbf{W}_{g^*} = \{i, j \in g\}, s_1 : n_g$$

$$\mathbf{W}_{g^*} = \begin{pmatrix} \mathbf{\Gamma}_{s_1} & 0 & \dots \\ 0 & \mathbf{\Gamma}_{s_2} & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}_{(n \times n)}$$

FIGURE I
Illustration of group matrix $\mathbf{W}_g$

$$\mathbf{W}_{g^*} = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$$

Weighting matrix $\mathbf{W}$ illustration for a network of 7 interacting firms in a group *pc8*. Of these, firms 1-4 share their postcode, as do firms 5-7.

I can extend this to more complicated set of group rules. I calculate non-overlapping group weights $\mathbf{W}_g$ to $\mathbf{W}_{g1}, \mathbf{W}_{g2}, \mathbf{W}_{g3}$ at once, which represent the successive concentric rings of *pc8* (core), *pc4*, and *ttwa*. The weighting rules for these multiple, non-overlapping groups is to zero the i, j interaction in W_{gk} if a prior matrix in the series, W_{gl} , $l < k$ has a non-zero entry in that cell. I illustrate an example in Figure II. These matrices are orthogonal.

$$w_{ij} \in \mathbf{W}_{gk} = \begin{cases} 0, & \text{if } i = j \\ 0, & \text{if } w_{ij,l} \neq 0, \exists l < k \\ \frac{1}{s_g - 1}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

Firms have been labelled using their location features to symmetric, transitive, non-overlapping groups by *ttwa* (largest) to *lat_lon5* (smallest), with the latter meaning two firm entries shared a latitude-longitude identifier to 5 decimal places. Reviewing the summary statistics, I opted for full postcodes, *pc8* as the group

TABLE V
Summary statistics across location groups

Property	#groups	Unclassified	Unclassified (%)	s_g avg. peers per group
<i>ttwa</i>	228	567	0.4%	668.3
<i>pc4</i>	2,705	0	0.0%	56.3
<i>pc8</i>	49,963	0	0.0%	3.1
<i>lat_lon3</i>	48,540	507	0.3%	3.1
<i>lat_lon4</i>	53,193	507	0.3%	2.9
<i>lat_lon5</i>	53,246	507	0.3%	2.9

FIGURE II
Illustration of multiple, non-overlapping groups

$$\mathbf{W}_{g1:3^*} \sim \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 & 0 \\ \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} & 0 & 0 & 0 \end{bmatrix}$$

Non-overlapping group illustration for a network of 8 interacting firms in a *ttwa*. Of these, firms 1-5 share the first half of a postcode - are in a *pc4* group, as are firms 6-8. Within those, firms 1-3 share a full postcode, as do firms 7-8. Firms 4-6 are isolated in their *pc8* groups.

identifier for initial 1-group models in Table ???. This was mostly driven by full data availability, and the consistency of postcode definitions with the larger group *pc4*. Deriving a subgroup by latitude-longitude match within the postcode seemed fruitless, as in many cases it would identify the exact same firms, a move from the average number of peers from 3.1 to 2.9, and was a similar number of groups, with non-cohesive definitions.

3.5. LOCATION DISTANCE CALCULATIONS

A network matrix is a more complex set of interactions where each individual i has their own set of peers by whom they are influenced. In the general definition (Bramoullé et al., 2009), the influence of each individual’s peer set is still the simple ‘leave-one-out’ average of their heterogenous peers, however, differing weights applied within the peer set are equally permissible. I use variants on the reciprocal of the spatial distance between the two firms to construct my weights in $\mathbf{W}_d$ in this analysis. Intuitively, these distance reciprocal weights are pretty similar. I explore 3 models just to demonstrate that findings are robust to small modifications in the weights, and that differing weighting strategies are permissible. A simple reciprocal weight is common to the gravity models used in the urban economics literature discussed¹⁸. A harsher penalty for distance, $\frac{1}{d_{ij}^2}$, has been employed by the New Economic Geography literature, emphasising a more crucial role for close proximity¹⁹.

$$w_{ijt} \in \Omega_t = \begin{cases} \frac{1}{d_{ij}}, & (1) \text{ gravity} \\ \frac{1}{d_{ij}^2} & (2) \text{ N.E.G.} \\ \exp -d_{ij} \cdot k & (3) \text{ exponential decline} \end{cases} \quad (2)$$

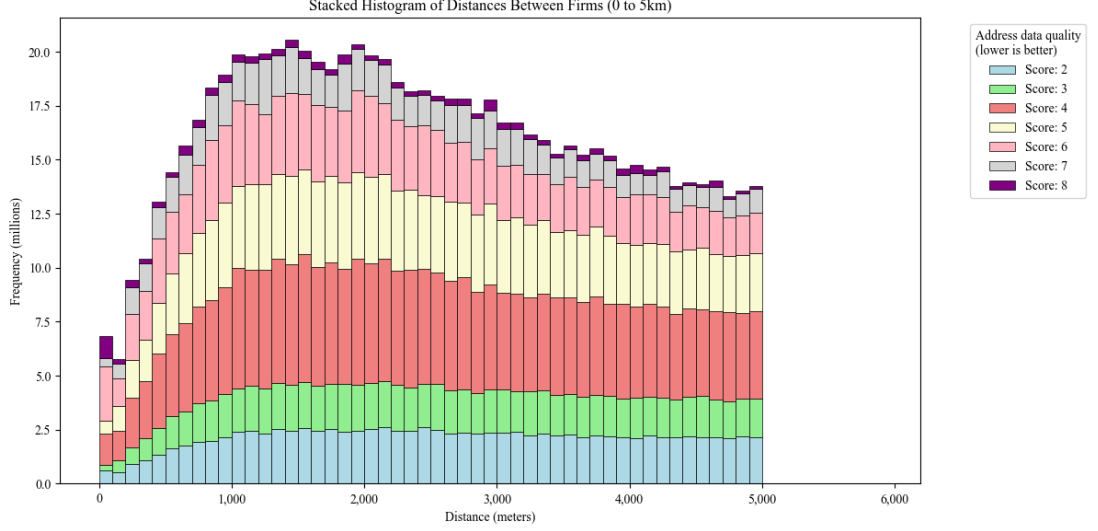
Distances are calculated as the Haversine distance between any 2 firms. This represents the straight-line, ‘as-the-crow-flies’, fastest path between the longitude and latitude pairs given, which has a slight adjustment from a standard straight-line distance due to the (great-circle) curvature of the Earth, for which the effects are minimal due to the small distances calculated. I show a histogram of distances calculated between firms in Figure III.

To reduce computational complexity, I disregard any possible matches which don’t share a Travel-to-Work-Areas (*ttwa*). Not only are these firms more likely to be further apart, eliminating the need to calculate distances between, say, London and Glasgow, the *ttwa* boundaries also a useful proxy for the functional boundaries of interacting economic agents, so it is a reasonable restriction to

¹⁸ (Gravity)

¹⁹ (NEG)

FIGURE III
Histogram of distances between any target-firm pairing



Plotted 0.1% sample of one-to-one relational table of distance calculations, into bins of 100m. Using the $d1$ distance measure, which is hard-bounded that no firm can have a firm-peer spillover relation past a $ttwa$ boundary, truncated at 5m. Colours indicate differing levels of data quality for addresses within Fame. Frequency in millions of target-firm connections.

limit the peer effects of firms across those borders. Additionally, I disregard any spatial connection over a 5km limit, saving on memory and storage. The threshold provides a plausible limit for the practical boundaries of one firm's peer effects on another, aligning helpfully with when contributions to a weighted average aggregate start to become too infinitesimal to matter using a reciprocal function for weights.

One final helpful property within the $\mathbf{W}_d$ matrix is the definition of a network itself. A network is the total set of individuals who have at least one interaction within the network:

$$j \in N_d \iff \mathbf{W}_{ij} \neq 0, \exists i \in N_d$$

There can exist several networks within a $\mathbf{W}$ interactions matrix, as certain individuals within a peer set may have these complex interactions, but all inside the set have nothing with those outside. Organising these networks in a $\mathbf{W}_d$ gives an outer block-diagonal structure, though with far larger blocks than $\mathbf{W}_g$. By disregarding all firm pairs which don't share a $ttwa$, I set the limits of a network to the $ttwa$ level. Networks offer further calculation opportunities for network fixed effects as discussed in section 6.3.

$\mathbf{W}_d$ is non-transitive. Firm B, a peer of firm A within its 5km radius, can clearly have its own peers that are within its own 5km radius that are beyond that radius for firm A. This matrix resembles a sparse matrix with a 'thick' diagonal,

rather than a matrix with distinct large blocks. I illustrate a network matrix using real sampled data from the cleaned dataset in Figure IV.

FIGURE IV
Illustration of distance-weighted network matrix $\mathbf{W}_d$

$$\mathbf{W}_{d^*} = \begin{bmatrix} 0 & 0.42 & 0.29 & 0.28 & 0 & 0 & 0 & 0 & 0 \\ 0.31 & 0 & 0.38 & 0.32 & 0 & 0 & 0 & 0 & 0 \\ 0.09 & 0.17 & 0 & 0.74 & 0 & 0 & 0 & 0 & 0 \\ 0.09 & 0.14 & 0.76 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.36 & 0.27 & 0.36 & 0.27 \\ 0 & 0 & 0 & 0 & 0.02 & 0 & 0.02 & 0.96 & 0.02 \\ 0 & 0 & 0 & 0 & 0.24 & 0.38 & 0 & 0.38 & 0 \\ 0 & 0 & 0 & 0 & 0.02 & 0.96 & 0.02 & 0 & 0.02 \\ 0 & 0 & 0 & 0 & 0.24 & 0.38 & 0 & 0.38 & 0 \end{bmatrix}$$

Illustrative 9×9 $\mathbf{W}$ network structure generated by taking a sample from the cleaned Fame data of a random block of 4 firms sharing a *ttwa*, and a random block of 5 firms. True $\mathbf{W}_d$ matrices have *ttwa* have much larger blocks within the block diagonal structure. Weights use the *d1* algorithm, which is the reciprocal of *distance_meters* row-normalised. Values rounded to 2 decimal places.

To calculate $\mathbf{W}_d$, I store distances d_{ij} in a one-to-one relational table, and compile rows on an efficient as-needed for calculations. The distance table has 734 million target-peer pairs; paradoxically, this implies both an incredibly rich, high-identification, interactions table due to a mean of 4,820 peers per target firm and also that the $\mathbf{W}_d$ matrix is mostly (over 96%) sparse. The *ttwa* and 5km boundaries are ultimately an arbitrary choice; stricter restrictions to create networks only within *pc8* boundaries would have resulted in a $\mathbf{W}_d$ matrix that is 99.8% sparse. While these could be considered reasonable restrictions, they matter a lot to the econometrics as firms disqualified as peers are not merely excluded from the weighted average calculations, but considered valid candidates for instruments, as they are non-transitive peers-of-peers. Errors in the specification of this parameter could lead to Incorrect specification of the peer group of the target firm. I discuss in section 6.2 how the most appropriate approach would be to cross-validate these non-linear parameter selections in a choice regression specification, minimising mean-squared-error (or adjusted R^2).

3.6. HIGHER-ORDER $\mathbf{W}$ TRANSFORMATIONS

Intuitively, I take the product of these weighting matrices $\mathbf{W}$ to the aggregate variables vectors *gva*, capital *total_assets*, and labour *employees*. As will be shown in section 4 ‘Static models’, the procedure to generate valid instrumental variables also requires me to apply multiples of this transformation matrix, exponents

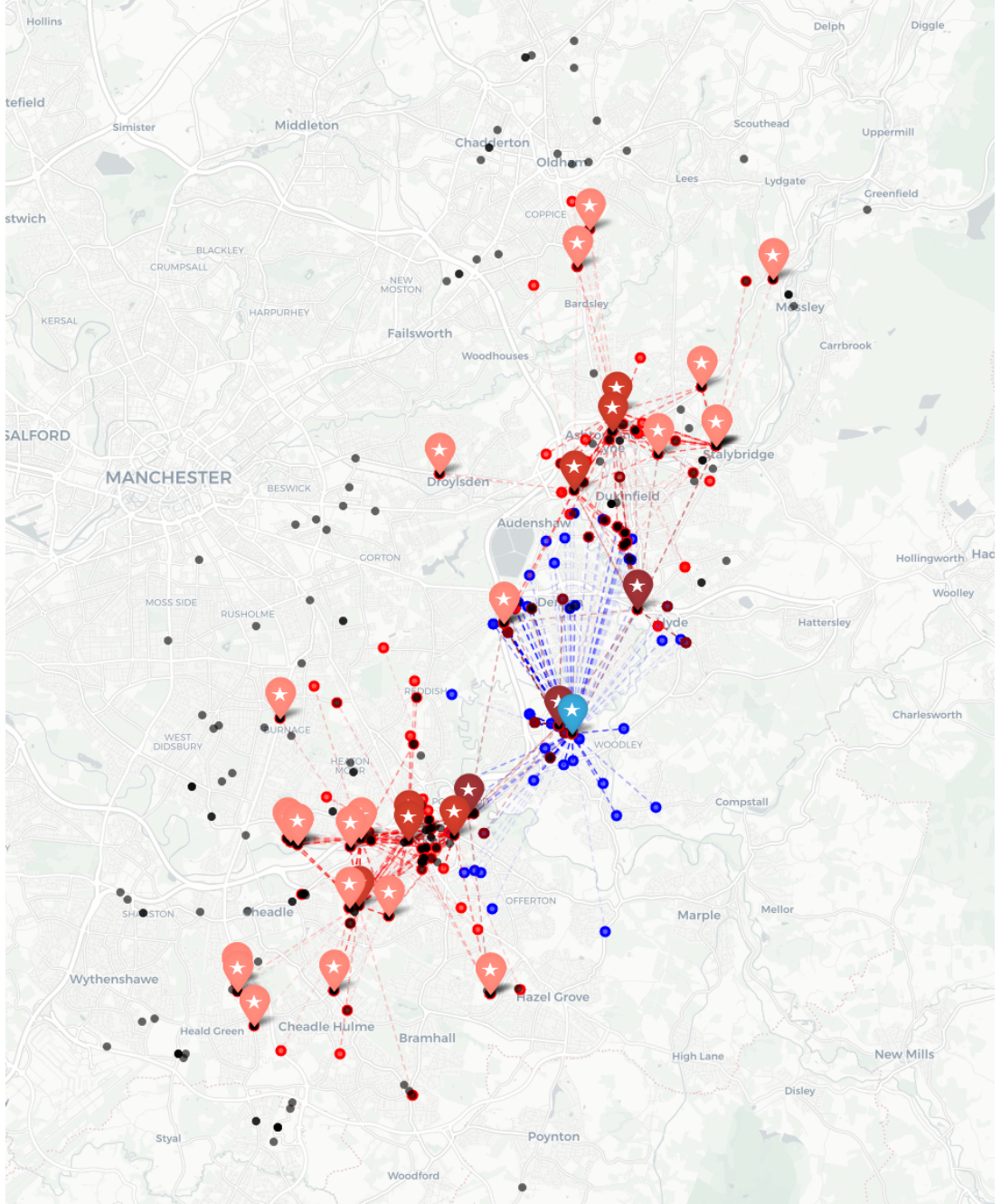
$\mathbf{W}^2, \mathbf{W}^3, \mathbf{W}^4$, and as an extension, the higher-order cross-transformation for group matrices, $\mathbf{W}_{g1} \mathbf{W}_{g2}, \mathbf{W}_{g1} \mathbf{W}_{g3}$ and so on to the aggregates.

I calculate these for my 3 aggregate variable vectors of interest. As given below in Table IX, this requires a total of 51 compound weighting matrices, or columns of a database, to be calculated and generated. Of course, these column vectors each contain a stacked firm (i)-year (t) observation, so the total computation is an additional matrix of 71 million entries. This applies similarly for distance matrices $\mathbf{W}_d$, and other modifications to any other procedures to generate weights that may be desired, as I discuss with in section 6.2 and Appendix C.

As reviewed in 2.4, the key condition for discovery of valid instruments in this peer effects context is the linear independence of $\mathbb{I}, \mathbf{W}, \mathbf{W}^2$. With the additional step of application of network fixed effects (section 59), this is additionally evaluated with $\mathbf{W}^3$. I can trivially say this is true, meeting the alternative sufficient conditions in I of ≥ 3 group sizes for the group $\mathbf{W}$ matrices.

My distance-decay weights, $\mathbf{W}_{d1}, \mathbf{W}_{d2}, \mathbf{W}_{d3}$ meet the strong alternative $s_d \geq 3$ condition, where s_d is the *diameter* of the network, the maximal number of links connecting any target i with another firm through a series of peers $j, k, \dots$. In the analysis of this set of firms with distance radius limits, the *diameter* is large, potentially spanning the full spatial diameter of a *ttwa*. It is this mechanism design provides another justification, alongside computation, for limiting the max length of a target-firm peer relation to 5km: it vastly increases the probability of ‘overlapping groups’, i.e. a high diameter network within a *ttwa*. The overlapping group network structure lends itself nicely to illustrations. Figure V selects a random firm i from my sample, which happens to be in Stockport, Greater Manchester, and draws the distances used to calculate metric $d1$ between this target and its peers. In typical network diagram fashion, a network with a considerable *diameter* is created by taking a subset of i ’s peers, then successively a set of their peers, then successively some of those and so on. 4 iterations in this diagram fails to clearly hit the *ittwa* boundary, on the south side, with more argument for this behaviour on the north side where peer nodes randomly selected may be starting to bunch at an ‘invisible’ border of the *ttwa* boundary.

FIGURE V
Illustration (2) of overlapping groups from a target firm to its whole network through successive peers



(blue pin): target firm i with a full set of its (blue dot) peers j , distances d_1 metric drawn in.
 (dark red pins): sample (limit 5) of j peers, then calculated as their own targets
 (red pins): sample (limit 5) of k peers of firms j , some of which are also a peer of i .
 (light red pins): sample (limit 5) of m peers of firms k , few of which are also a peer of i .
 (light red dots): set of n peers of firms m , few of these are peers of i, j , probabilistically creating a network diameter of $s_d \geq 3$. Plotting software from Folium (2012). Map tiles from CARTO (2026).

4. STATIC MODELS

I use a stylised 3-factor production model throughout this analysis. Y , A , K , and L are each firm-year observation's GVA, TFP, labour, and capital in levels. I denote logged terms with lower-case notation and vectors in bold.

Log TFP, denoted by $z_{i,t}$, is the residual after the own-firm contribution of the two classical inputs. I decompose it to a constant across all observations, firm and time fixed effects and, most crucially for this investigation, a vector of the weighted sum of all peer 3-factor production factors with the dot product against the vector of their respective spillover coefficients $\boldsymbol{\theta}_0$:

$$\mathbf{z}_t^{spill} = \left(\sum_{j \neq i} w_{ij} z_{j,t} \quad \sum_{j \neq i} w_{ij} k_{j,t} \quad \sum_{j \neq i} w_{ij} l_{j,t} \right) \quad (3)$$

$$\boldsymbol{\theta}_0 = \begin{pmatrix} \delta_0 & \xi_0 & \zeta_0 \end{pmatrix}' \quad (4)$$

Different static weights, w_{ij} , are explored throughout this section. The objective is to estimate the vector of structural coefficients $\boldsymbol{\theta}_0$ of spillover from peers to any given target firm.

$$Y_{i,t} = A_{i,t} \cdot K_{i,t}^{\beta_1} \cdot L_{i,t}^{\beta_2} \quad (5)$$

$$z_{i,t} = \ln(A_{i,t}) = y_{i,t} - \beta_1 k_{i,t} - \beta_2 l_{i,t} \quad (6)$$

$$= \alpha_i + \gamma_t + \beta_0 + \mathbf{z}_t^{spill} \boldsymbol{\theta}_0 + \varepsilon_{i,t} \quad (7)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} \quad (8)$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \xi_0 \sum_{j \neq i} w_{ij} k_{j,t} + \zeta_0 \sum_{j \neq i} w_{ij} l_{j,t} + \varepsilon_{i,t}$$

$$= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t}$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} \cdot (y_{j,t} - \beta_1 k_{j,t} - \beta_2 l_{j,t}) + \xi_0 \sum_{j \neq i} w_{ij} k_{j,t} + \zeta_0 \cdot \sum_{j \neq i} w_{ij} l_{j,t}$$

$$= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} \quad (9)$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t}$$

Equation (9) is a linear peer effects specification as given by Manski (1993). It is helpful to borrow the terminology from this discussed in section 2.4 and keep in mind the conceptual distinction between these and the more general and related concepts of endogeneity, exogeneity, and correlation. Variables $(k_{i,t} \quad l_{i,t}) \in x_{i,t}$ can be thought of as *exogenous* characteristics of firm i causally affecting its outcome $y_{i,t}$. There is an *endogenous* effect directly from peer outcomes $y_{j,t}$ with coefficient δ_0 . Peers also have *exogenous* characteristics $x_{i,t}$, which have a direct

exogenous effect on the target’s productivity $x_{i,t}$ via ξ_0 and ζ_0 . For this section of models, I assume there are no *correlated* effects and leave further discussion of the implementation of these to subsection 6.3 ‘Network fixed effects’. The use of panel fixed effects has no effect on econometrics discussed in section 2.4.

It is important to note that I do not impose restrictions on θ_0 . I allow each of the 3 factors of production to independently exert a spillover effect in general form. This allows this analysis’ investigation as to which of the production factors matters more for productivity spillovers, and in what ratio, conceptually linked to the discussion in 2.1 about identifying the channels through which productivity spillovers might occur. Through this 3-factor production model, it is hard to link each parameter in θ_0 to a Marshallian channel, but broadly we might consider ζ_0 , the coefficient on an increase in a peer firm’s labour inputs affecting a target’s TFP to resemble Marshall (1920)’s labour pooling, with some interaction to knowledge spillovers, and δ_0 , the coefficient from a peer firm’s TFP to target TFP to be the remainder of knowledge spillover effects. A TFP spillover occurring from peer firm’s capital ξ_0 would be a novel channel to Marshall (1920)’s discussion and could reflect the idea of more capital-intensive peers. Regardless, it is strongly justifiable that we would expect the coefficient on each peer production factor to spillover to a target firm’s production heterogeneously, and not be proportionate to the contribution of these factors to their own-firm GVA. I have no *a priori* arguments for the sign of the θ_0 coefficients and their relative magnitudes compared to their contributions to own-firm GVA in β_1, β_2 .

$$y_{i,t}^{(y)} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + \varepsilon_{i,t} \quad (10)$$

$$y_{i,t}^{(z)} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} z_{j,t} + \varepsilon_{i,t} \quad (11)$$

$$= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} - \delta_0 \beta_1 \sum_{j \neq i} w_{ij} k_{j,t} - \delta_0 \beta_2 \sum_{j \neq i} w_{ij} l_{j,t}$$

Two alternative specifications may be proposed: equation (10) modifies (8) to argue that it is the whole of GVA which matters the peer-target spillover, not each component heterogeneously. It is trivial to see that this implies the coefficient on the *exogenous effect* is zero, such that $\xi - \delta_0 \beta_1 = \zeta - \delta_0 \beta_2 = 0$. This has the interpretation alluded to above, that the peer effect of any classical factor, capital and labour, is exactly in the ratio to its own factor income, which is a strong restriction to impose. We might expect this to be a nested specification of (9), that we can merely Wald-test the joint hypothesis as given above, but actually this specification cannot be identified due to the endogeneity of $y_{j,t}$, without strict assumptions on the orthogonality of peer errors.

The alternative in equation (11) argues that only the TFP component of production matters for the spillover to influence a target firm's TFP. This is also a restriction on the more general (8) such that $\xi_0 = \zeta_0 = 0$. At first glance, it may appear that this model follows the structure of the peer effect class of models too and is in fact over-identified, with both the coefficients of the peer classical factors and $\sum_{j \neq i} w_{ij} y_{j,t}$ identifying δ_0 . However, because of the specific set-up of the reflection problem where $w_{ij} y_{j,t}$ is endogenous and we instrument it through $k_{j,t}, l_{j,t}$ higher-order spatial lags, this model is actually non-identified. I demonstrate this in the next section 4.1.

While the classical production function (8) is limited to constant returns to scale, $\beta_1 + \beta_2 \leq 0$, a standard assumption across the growth and firm production literature, no such restrictions for the firm occur with the endogenous decomposition of TFP. A restriction on logged production (8) collectively on β_1, β_2 , and θ_0 is difficult to justify. As with other external effects, there is no reason to believe that market factor prices and the firm decision-rule for spatial allocation in view of these spillover effects, $w_{ij} z_{j,t}$, is socially optimal. A peer firm j with extremely high and growing components of GVA has little interest in siting themselves in the centre of a cluster of firms that are struggling with TFP growth, yet the external impacts of that firm would be significant.

4.1. STATIC PANEL IDENTIFICATION

Consistency requires the plausible exogeneity of regressors used in structural equation (9) from the error term $\varepsilon_{i,t}$. Capital and labour inputs in every firm i, j are assumed to be strictly exogenous to every other firm's TFP error shock, for every period in this static model. This is crucial for consistency with the direct own-firm $k_{i,t}, l_{i,t}$ inputs, but also the use of $\mathbf{w}_{ij} k_t, \mathbf{W}^k l_t$ as an *exogenous* effect and, as we will later see, the ability to leverage higher-order spatial lags of the *exogenous* effect as an unconfounded instrument for $w_{ij} y_{j,t}$. This is a restrictive assumption, but standard in the practice of constructing static panel models. It is relaxed in Section 5.

$$\mathbb{E}[\varepsilon_{i,t} | (k_{i,t} \quad l_{i,t})] = 0_{(1 \times 2)}, \quad \forall i, \forall t, \tau \quad (12)$$

The *endogenous* effect regressor $\sum_{j \neq i} w_{ij} y_{j,t}$ is, unsurprisingly, endogenous, making valid estimation in this setting more complicated. To show the endogeneity requires stacking w_{ij} target-peer weights to an $n \times n$ interactions matrix, $\mathbf{W}$. I invert the *endogenous* component to give a $(\mathbb{I} - \delta_0 \mathbf{W})^{-1}$ coefficient on all variables.

$$\delta_0 < 1 \implies |\mathbb{I} - \delta_0 \mathbf{W}| \neq 0 \quad (13)$$

The inversion is permissible because I can expect a restriction to apply, ruling out iteratively increasing shocks²⁰. We can interpret this as a ‘spatial lag polynomial’, directly comparable to the lag polynomial and MA inversion procedure in autoregressive time series analysis²¹.

$$\mathbf{y}_t = \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \delta_0 \mathbf{W} \mathbf{y}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \quad (14)$$

$$\begin{aligned} &= (\mathbb{I} - \delta_0 \mathbf{W})^{-1} \left(\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \right. \\ &\quad \left. + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \right) \\ &= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \left(\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \right. \\ &\quad \left. + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \right) \end{aligned} \quad (15)$$

$$\begin{aligned} &= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \boldsymbol{\alpha} + \frac{\gamma_t + \beta_0}{1 - \delta_0} \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \\ &\quad + \xi_0 \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^{k+1} \mathbf{k}_t + \zeta_0 \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^{k+1} \mathbf{l}_t + \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \boldsymbol{\varepsilon}_t \end{aligned} \quad (16)$$

$$y_{j,t} = \cdots + \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^{k+1}]_{(j, \cdot)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_0 \\ \zeta_0 \end{pmatrix}}_{\text{exogenous}_j} + \underbrace{\sum_{k=0}^{\infty} [\delta_0^k \mathbf{W}^k]_{(j, \cdot)} \cdot \boldsymbol{\varepsilon}_t}_{\text{error transformation}_j} \quad (17)$$

It is the primary econometric challenge of this section to overcome the endogeneity of $y_{j,t}$, which is clearly demonstrated in reduced-form equation (17). It contains a linear transformation of all firms’ error shocks $\boldsymbol{\varepsilon}_t$, including $\varepsilon_{i,t}$. This implies:

$$\mathbb{E}[y_{j,t} \varepsilon_{i,t}] \neq 0 \quad (18)$$

The ‘Reflection Problem’ poses a second issue as outlined in Manski (1993). Equation (16) shows that $\mathbf{W} \mathbf{y}_t$ is a linear combination of all other regressors. In the classical setup, $\mathbf{W}$ is a block diagonal matrix of equal-size groups, and Manski (1993), later reaffirmed in Bramoullé et al. (2009) show that the reduced-form estimation of this model contains fewer estimated parameters than structural

²⁰ $\delta_0 \geq 1$ would imply an exogenous TFP shock to firm k would increase TFP in its peer firm j more than 1x, which would increase TFP in that firm’s peer i more than 1x, and so on until the effect multiplies to infinity, or at least on the knife-edge $\delta_0 = 1$ case, any TFP shock occurring in one firm applies equally and consistently to every firm in a network. Estimating δ_0 is a helpful indicator of a multicollinear model with unstable coefficient estimates.

²¹ Note that vectors $\mathbf{k}_t, \mathbf{l}_t$ appear twice in (16), as components of $\mathbf{y}_t$ so coefficients can be grouped but it is clearer to separate this into an ‘own-firm’ effect of $\mathbf{k}_t, \mathbf{l}_t$ and a spillover effect of $\mathbf{W}^{k+1} \mathbf{k}_t, \mathbf{W}^{k+1} \mathbf{l}_t \forall k \geq 0$

parameters, making variation in $y_{j,t}$ indistinguishable to that of all other regressors.

There are two solutions. I could either proceed with an extremely strong exogeneity assumption, (19), which is most likely satisfied by total orthogonality between all contemporaneous idiosyncratic errors hitting peer firms (20). This in turn makes the regressor $y_{j,t}$ exogenous, allowing direct estimation of (8), sidestepping the ‘Reflection Problem’ by assuming away *endogenous* regressors, and having each regressor exogenously identified in itself. For identification, this implies that any productivity shock $\varepsilon_{j,t}$ occurring in peer firm j , if it is contemporaneously correlated with TFP $z_{i,t}$ in target firm i must be attributable to the peer effect coefficient via δ_0 . This precludes any contemporaneous, correlated independent TFP shocks, which might occur on both the global and local level. Model (1.) takes this assumption in 4.2. We should assess these results with the heavy caveat that this strict error orthogonality assumption is unlikely to be credible.

$$\begin{aligned}\mathbb{E}[\varepsilon_i | \mathbf{W}^k (\boldsymbol{\alpha} \quad \gamma_t \boldsymbol{\iota} \quad \beta_0 \boldsymbol{\iota} \quad \mathbf{k}_t \quad \mathbf{l}_t)] &= 0, \quad \forall k \\ \mathbb{E}[\varepsilon_{i,t} | \mathbf{W}^k \boldsymbol{\varepsilon}_t] &= 0, \quad \forall i, j \\ \iff \mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] &= 0, \quad \forall k\end{aligned}\tag{19}$$

Instead, as recommended in Bramoullé et al. (2009), the most defensible approach is to internal instrument the *endogenous* variable, $y_{j,t}$ through deeper spatial lags of the *exogenous* parameters $\mathbf{W}^2 \mathbf{x}_t, \mathbf{W}^3 \mathbf{x}_t \dots \mathbf{W}^k \mathbf{x}_t \quad \forall k > 2$. These are valid instruments under certain testable properties of $\mathbf{W}$ given in Table I. Essentially, these reduce to checking that the matrices $\mathbb{I}, \mathbf{W}, \mathbf{W}^2$ are linearly independent, which are sufficiently achieved for group interactions by having 2 or more differently-sized groups, or a network diameter of at least 2.

This identification strategy is not a ‘free lunch’. It requires the underlying assumption that the group size, or network diameter, is in itself exogenously determined. This would be intuitively violated, if firms colocate based on the expected productivity benefits of moving to an area, namely to benefit from the supply-side spillovers we aim to investigate. Unfortunately, it is highly likely that firm location decisions are endogenous, this is the whole principle of the (New Economic Geography) literature whose descriptive patterns of firm movements underlying productivity spillover I here aim to investigate. I am helped here, however, by the use of time-invariant weights. Plausibly, over the observations of a panel dataset up to 20 years, if firm weights apply to each firm on a fixed basis, the temporal variation despite these constant location effects provide credible exogenous variation. On this however, the strategy runs into the limitations of Fame location data discussed in 3.3. The analysis is also helped by the relatively

higher number of panel observations to relatively low ratio of confounding movers in the dataset, as well as the numerous, plausibly exogenous reasons why a firm might choose to set up in a location with path dependency, rather than to fully maximise its peer spillover benefits. On the balance of arguments, I take the assumption of network exogeneity and identification approach for the group models in Table VI.

I use the instruments $\mathbf{W}^2\mathbf{x}_t$ and other second-order variants for this section's group models VI and $\mathbf{W}^2\mathbf{x}_t, \mathbf{W}^3\mathbf{x}_t, \mathbf{W}^4$ as feasible instruments for this section's network models in Table VII. Equation (21) illustrates the valid $2l$ moment conditions, noting that I have chosen to display this in a $l \times 2$ matrix for space and conceptual symmetry, as the instruments can be grouped by k_t and l_t regressors. The first 2 rows correspond to the exogenous regressors within the reduced-form equation itself, the own-firm classical factors effect from $k_{i,t}, l_{i,t}$ and the *exogenous* peer effect from $k_{j,t}, l_{j,t}$. The remaining rows are instrument linear combinations of all firm classical factor inputs $\mathbf{k}_t, \mathbf{l}_t$, using the higher-order spatial lags of $\mathbf{W}$ discussed.

$$\mathbb{E}[\epsilon_{i,t} \mid \begin{pmatrix} k_{i,t} & l_{i,t} \\ w_{ij}k_{j,t} & w_{ij}l_{j,t} \\ [\mathbf{W}^2]_{(j,)} \mathbf{k}_t & [\mathbf{W}^2]_{(j,)} \mathbf{l}_t \\ [\mathbf{W}^3]_{(j,)} \mathbf{k}_t & [\mathbf{W}^3]_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \\ [\mathbf{W}^k]_{(j,)} \mathbf{k}_t & [\mathbf{W}^k]_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{l \times 2} \quad (21)$$

These instruments and conditions are intuitive: we can see through equation (17) that a linear combination of all firms' contemporaneous capital and labour inputs, $\mathbf{W}^{k+1}(\mathbf{k}_t \ \mathbf{l}_t) \ \forall k \geq 0$ are an exogenous component of $y_{j,t}$ with the coefficients ξ_0 and ζ_0 respectively. It has relevance, meaning $\mathbb{E}[y_{j,t} \cdot [\mathbf{W}^k]_{(j,)}\mathbf{x}_t] \neq 0, \ \forall k > 2$ as long as $\xi_0 \neq 0$ and $\zeta_0 \neq 0$, which occurs in all unrestricted cases distinct from (11). The relevance conditions themselves are also unsurprising and highly intuitive given the model's setup with GVA equation (8). Note that the relevance conditions are more complex setup in Bramoullé et al. (2009), this is because the factor $\delta_0(\beta_1 \ \beta_2)$ cancels out due the accounting composition of TFP in the *exogenous effect*, which is absent from the general model. This also allows clear understanding of why the TFP-only spillover model (11) fails, as setting both $\xi_0 = \zeta_0 = 0$ means the higher order spatial lags of $\mathbf{k}_t$ and $\mathbf{l}_t$ disappears from the composition of the $y_{j,t}$ term. An Anderson-Rubin test for weak instruments in my nested 3-factor spillover setup can test for the strength of instrument of $\mathbf{k}_t$ and $\mathbf{l}_t$. If I find that the higher-order spatial lags are jointly weak instruments,

that would validate the TFP-only spillover model and suggest $\xi_0 = \zeta_0 = 0$.

Finally, one might observe that $[\mathbf{W}]_{(j)}\mathbf{x}_t$ is a component of $y_{j,t}$ and so a valid instrument. However, it does not increase the number of instruments I would use in my instrumental specification as it is already included as an exogenous regressor in 14 so does not help increase our identification of $y_{j,t}$.

4.2. LINEAR-IN-MEANS GROUPED MODEL

The initial model proceeds with the strict error orthogonality (19) and (20) assumptions. This is helpful as a benchmark to compare the IV strategy, with more credible assumptions, that I then adopt. Model (1.) estimates (22), noting that the single group interactions matrix $\mathbf{W}_g$ corresponds to the *pc8* group.

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} \quad (\text{from 9})$$

$$+ \delta_0 \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij,g} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij,g} l_{j,t}$$

$$y_{i,t} = \alpha_i + \gamma_t + \pi_0 + \pi_1 k_{i,t} + \pi_2 l_{i,t} \quad (22)$$

$$+ \pi_3 \sum_{j \neq i} w_{ij,g} y_{j,t} + \pi_4 \sum_{j \neq i} w_{ij,g} k_{j,t} + \pi_5 \sum_{j \neq i} w_{ij,g} l_{j,t} + \varepsilon_{i,t}$$

$$\mathbb{E}[\varepsilon_{i,t} \cdot (k_{i,t} \quad l_{i,t} \quad y_{j,t} \quad k_{j,t} \quad l_{j,t})'] = 0_{(5 \times 1)} \quad (23)$$

$$\begin{pmatrix} \hat{\pi}_0 \\ \hat{\pi}_1 \\ \hat{\pi}_2 \\ \hat{\pi}_3 \\ \hat{\pi}_4 \\ \hat{\pi}_5 \end{pmatrix} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \delta_0 \\ \xi_0 - \delta_0 \beta_1 \\ \zeta_0 - \delta_0 \beta_2 \end{pmatrix} \quad (24)$$

With 5 structural and 5 reduced-form parameters, this model is identified. The initial static model applies simple time-invariant weights w_{ij} in a group structure, $\mathbf{W}_{g1}$ I discussed group assignment in section 3.5.

Model (2) instruments these groups via $\mathbf{W}^2 \mathbf{x}$, The reduced-form equation is given in (25). The IV strategy has the moment conditions in (21), and estimates parameters using continuously-updating Generalised Method of Moments (GMM-CUE), of calculating an efficient transformation matrix to weigh moment conditions (21) to improve the estimator's efficiency. All the IV models in this analysis are overidentified, as helpfully the structural model has 2 *exogenous* parameters in capital and labour for 1 *endogenous* GVA term. In the group model, I present additional instruments up to the 3rd-order and 4th-order lags, as the first-stage 2SLS results presented in Appendix B Table X suggests a considerable

jump in adjusted-R² when moving from 3rd-order to 4th-order lag instruments.

$$\begin{aligned}
y_{i,t} &= \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \varepsilon_{i,t} & (\text{from 9}) \\
&+ \delta_0 \sum_{j \neq i} w_{ij,g} y_{j,t} + (\xi_0 - \delta_0 \beta_1) \cdot \sum_{j \neq i} w_{ij,g} k_{j,t} + (\zeta_0 - \delta_0 \beta_2) \cdot \sum_{j \neq i} w_{ij,g} l_{j,t} \\
\Delta y_{i,t} &= \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} & (25) \\
&+ \beta_3 \Delta \sum_{j \neq i} w_{ij,g} y_{j,t} + \beta_4 \Delta \sum_{j \neq i} w_{ij,g} k_{j,t} + \beta_5 \Delta \sum_{j \neq i} w_{ij,g} l_{j,t} + \Delta \varepsilon_{i,t} \\
\mathbb{E}[\epsilon_{i,t} \mid &\begin{pmatrix} k_{i,t} & l_{i,t} \\ w_{ij} k_{j,t} & w_{ij} l_{j,t} \\ \begin{bmatrix} \mathbf{W}_g^2 \end{bmatrix}_{(j,)} \mathbf{k}_t & \begin{bmatrix} \mathbf{W}_g^2 \end{bmatrix}_{(j,)} \mathbf{l}_t \\ \begin{bmatrix} \mathbf{W}_g^3 \end{bmatrix}_{(j,)} \mathbf{k}_t & \begin{bmatrix} \mathbf{W}_g^3 \end{bmatrix}_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \\ \begin{bmatrix} \mathbf{W}_g^k \end{bmatrix}_{(j,)} \mathbf{k}_t & \begin{bmatrix} \mathbf{W}_g^k \end{bmatrix}_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{l \times 2} & (\text{from 21})
\end{aligned}$$

Results for Model (1.) and (2.) are given in Table VI.

As a brief extension, I explore the possibility of regressing multiple, non-overlapping spatial groups at once. This is a natural examination of the specification as I already have the $\mathbf{W}_{g1}$, $\mathbf{W}_{g2}$, $\mathbf{W}_{g3}$ concentric doughnut group transformations for *pc8*, *pc4* and *ttwa* to run all in the panel IV model at once. The algebraic derivation of endogeneity and valid instruments is very similar to those in the single-regressor problem, where the inversion term is the sum matrices weighted by their coefficient δ_s .

TABLE VI
LMM: 1 peer regressor with group weights, error
orthogonality model vs instrumented

	(1.) exog	(2.) reduced-iv	(3.) iv4
β_1	0.261 (0.005)***	0.233 (0.008)***	0.234 (0.008)***
β_2	0.650 (0.005)***	0.555 (0.007)***	0.555 (0.007)***
δ_0	0.199 (0.004)***	-0.384 (0.177)**	-0.414 (0.177)**
ξ_0	-0.005 (0.002)**	0.029 (0.014)**	0.031 (0.014)**
ζ_0	0.011 (0.003)***	0.035 (0.013)***	0.037 (0.013)***
J-test		X (0.330)	X (0.357)
Instr.		$\mathbf{W}_{g1}^2 \mathbf{k}, \mathbf{W}_{g1}^2 \mathbf{l}$ $\mathbf{W}_{g1}^3 \mathbf{k}, \mathbf{W}_{g1}^3 \mathbf{l}$	$\mathbf{W}_{g1}^2 \mathbf{k}, \mathbf{W}_{g1}^2 \mathbf{l}$ $\mathbf{W}_{g1}^3 \mathbf{k}, \mathbf{W}_{g1}^3 \mathbf{l}$ $\mathbf{W}_{g1}^4 \mathbf{k}, \mathbf{W}_{g1}^4 \mathbf{l}$
Obs.	726,283	587,772	587,772
R ² (excl. F.E.)	0.3550	0.0012	-0.0139
R ² (Overall)	0.7966	0.0000	0.0000

Regression tables given structural parameters calculated using transformations of the reduced-form parameters. Stars denote significance against an H_0 with the usual confidence levels, but these are not meaningful for this analysis due to causal interpretations discussed.

$$\mathbf{y}_t = \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t \quad (26)$$

$$\begin{aligned} & + \sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \mathbf{y}_t + \sum_{s=1}^3 (\xi_s - \delta_s \beta_1) \mathbf{W}_{gs} \mathbf{k}_t + \sum_{s=1}^3 (\zeta_s - \delta_s \beta_2) \mathbf{W}_{gs} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \\ & = (\mathbb{I} - \sum_{s=1}^3 \delta_s \mathbf{W}_{gs})^{-1} \left(\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \boldsymbol{\varepsilon}_t \right. \\ & \quad \left. + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t \right) \\ & = \sum_{k=0}^{\infty} \left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \boldsymbol{\alpha} + \frac{\gamma_t + \beta_0}{1 - \sum_{s=1}^3 \delta_s} \boldsymbol{\iota} + \sum_{k=0}^{\infty} \left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k (\beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t) \quad (27) \\ & \quad + \sum_{k=0}^{\infty} \left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \sum_{s=1}^3 \mathbf{W}_{gs} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_s - \delta_s \beta_1 \\ \zeta_s - \delta_s \beta_2 \end{pmatrix} + \sum_{k=0}^{\infty} \left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \boldsymbol{\varepsilon}_t \end{aligned}$$

$$\begin{aligned} y_{j,t} = & \cdots + \underbrace{\sum_{k=0}^{\infty} \left[\left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \sum_{s=1}^3 \mathbf{W}_{gs} \right]_{(j, \cdot)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_s - \delta_s \beta_1 \\ \zeta_s - \delta_s \beta_2 \end{pmatrix}}_{\text{exogenous}_j} \\ & + \sum_{k=0}^{\infty} \left[\left(\sum_{s=1}^3 \delta_s \mathbf{W}_{gs} \right)^k \right]_{(j, \cdot)} \cdot \boldsymbol{\varepsilon}_t \end{aligned} \quad (28)$$

This strategy of multiple weighting groups at once opens up interesting additional cross-transformation instruments. These cross-transformations are possible and create new, linearly independent regressors as long as there is sufficient heterogeneity between the weighting matrices, which here is theoretically guaranteed by the mutual orthogonality of $\mathbf{W}_g$. Any combination of higher-order $\mathbf{W}_{gs}$ group transformations can be employed as instruments (29), or the linear composite of the 3 group transformation matrices, which intuitively represents all firms in a *ttwa* can be used in higher orders (30).

$$\mathbb{E}[\epsilon_{i,t} \mid \begin{pmatrix} [\mathbf{W}_{g1} \mathbf{W}_{g2}]_{(j, \cdot)} \mathbf{k}_t & [\mathbf{W}_{g1} \mathbf{W}_{g2}]_{(j, \cdot)} \mathbf{l}_t \\ [\mathbf{W}_{g2} \mathbf{W}_{g3}]_{(j, \cdot)} \mathbf{k}_t & [\mathbf{W}_{g2} \mathbf{W}_{g3}]_{(j, \cdot)} \mathbf{l}_t \\ \vdots & \vdots \end{pmatrix}] = 0 \quad (29)$$

$$\mathbb{E}[\epsilon_{i,t} \mid \begin{pmatrix} \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g1} \right]_{(j, \cdot)} \mathbf{k}_t & \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g1} \right]_{(j, \cdot)} \mathbf{l}_t \\ \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g2} \right]_{(j, \cdot)} \mathbf{k}_t & \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g2} \right]_{(j, \cdot)} \mathbf{l}_t \\ \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g3} \right]_{(j, \cdot)} \mathbf{k}_t & \left[\sum_{s=1}^3 \mathbf{W}_{gs} \mathbf{W}_{g3} \right]_{(j, \cdot)} \mathbf{l}_t \end{pmatrix}] = 0_{3 \times 2} \quad (30)$$

However, in practice with this dataset, regressing more groups has proved unlikely to offer more insights without additional diagnosis. Coefficients reported in

Appendix B Table XI appear unstable, though this is slightly alleviated by omitting the outer *ttwa* group and regressing only using *pc8* and *pc4* $\mathbf{W}_g$ transformations. Nevertheless, in the results given here, these fail the standard Sargan-Hansen test for overidentifying restrictions. In addition, employing more instruments, the higher-order cross-transformations of $\mathbf{W}_{gs}$ discussed is theoretically sound, but led to multicollinearity in this dataset through any approach. This could be due to the fact that each $\mathbf{W}_{gs}$ is not truly independent and orthogonal, the identities of *pc8* and *pc4* are linked.

This extension is interesting and useful, because the assignment of groups, especially through more sophisticated dimensions as discussed in section 6.2 could lead to better understanding of firm's heterogenous peer groups, for example, through industry or regional cuts. Perhaps these will have more success by producing independent, orthogonal matrices than the multiple orthogonal groups extension run here.

4.3. STATIC DISTANCE DECAY

I employ a similar static specification strategy, but employing $\mathbf{W}_d$ distance-decayed weights in a network structure, rather than group weights. The reciprocal distance weights are stylistic of a declining peer influence further away, representing a lower frequency of social collisions that might inspire knowledge spillovers, barriers created via transport and time costs, and decreased ease of peer observation. Similarly, if this analysis' hypothesis is that spatial proximity is key to determination of a peer group, it is intuitive that increasing distance, as a continuous measure, would imply a decline in influence, rather than equal influence of all peers in an arbitrary cut-off.

I estimate this model employing the spatial lag instruments only, dropping the error orthogonality variant. The reduced-form equation and moment conditions of these models are identical to those of the group setting, except for the use of different weights which show up in $w_{ij,d}$ and $\mathbf{W}_d$.

$$\begin{aligned} \Delta y_{i,t} = & \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} \\ & + \beta_3 \Delta \sum_{j \neq i} w_{ij,d} y_{j,t} + \beta_4 \Delta \sum_{j \neq i} w_{ij,d} k_{j,t} + \beta_5 \Delta \sum_{j \neq i} w_{ij,d} l_{j,t} + \Delta \varepsilon_{i,t} \end{aligned} \quad (31)$$

$$\mathbb{E}[\varepsilon_{i,t} \mid \begin{pmatrix} k_{i,t} & l_{i,t} \\ w_{ij} k_{j,t} & w_{ij,d} l_{j,t} \\ [\mathbf{W}_d^2]_{(j,)} \mathbf{k}_t & [\mathbf{W}_d^2]_{(j,)} \mathbf{l}_t \\ [\mathbf{W}_d^3]_{(j,)} \mathbf{k}_t & [\mathbf{W}_d^3]_{(j,)} \mathbf{l}_t \\ \vdots & \vdots \\ [\mathbf{W}_d^k]_{(j,)} \mathbf{k}_t & [\mathbf{W}_d^k]_{(j,)} \mathbf{l}_t \end{pmatrix}] = 0_{l \times 2} \quad (\text{from ??})$$

Results for the 3 respective reciprocal distance measures are given in Table VII. Note that each of the distance d_1, d_2, d_3 measures are non-orthogonal, distinct transformations and so the properties discussed for the W_{gs} matrices allowing them to be regressed at once do not here apply. In addition, I include a specification exploring employing more lags for overidentification.

Model (4.) includes more lags which improves significance. The inclusion of more lags continues to pass the Hansen J-test, avoiding over-specification which would be tested by a rejection of the null hypothesis. I include an additional table in Appendix ?? which shows the first-stage Panel OLS estimates of the endogenous regressor $\mathbf{W}_y$ using the instruments employed, were this specification to be ran through 2SLS²². This shows improvements in adjusted R^2 for the model up to 3 lags, but barely any improvement for including an additional 4th-order

²² Noting that in practice, I use GMM-CUE for efficiency of estimates

TABLE VII
Static panel IV using continuous distance-decay $\mathbf{W}_d$

	(1.) dd1	(2.) dd2	(3.) dd3	(4.) dd1-deep3
β_1	0.271 (0.007)***	0.271 (0.007)***	0.271 (0.007)***	0.267 (0.007)***
β_2	0.512 (0.006)***	0.512 (0.006)***	0.512 (0.006)***	0.513 (0.006)***
δ_0	-0.159 (0.286)	-0.110 (0.233)	-0.353 (0.309)	-0.740 (0.388)*
ξ_0	0.011 (0.019)	0.008 (0.017)	0.033 (0.030)	0.052 (0.029)*
ζ_0	0.023 (0.037)	0.015 (0.027)	0.049 (0.041)	0.098 (0.050)*
J-test	X (0.311)	X (0.989)	X (0.969)	X (0.100)
Instr.	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$	$\mathbf{W}_{d2}^2 \mathbf{k}, \mathbf{W}_{d2}^2 \mathbf{l}$	$\mathbf{W}_{d3}^2 \mathbf{k}, \mathbf{W}_{d3}^2 \mathbf{l}$	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$ $\mathbf{W}_{d1}^3 \mathbf{k}, \mathbf{W}_{d1}^3 \mathbf{l}$
Obs.	854,192	854,192	854,192	854,192
R ²	0.0985	0.1034	0.0980	-0.1213

lag.

To investigate this issue, in Appendix ?? Table XIII I run an extension where firms which share a location, meaning distance is mechanically 0, are excluded. Results are reported in ?. On evaluation by R^2 these models perform slightly worse. An odd phenomenon is that the signs flip on the spillover parameters. These specifications reveal an emerging feature of these models, which is that $\delta_0 < 0 \implies \xi_0 > 0 \cap \zeta_0 > 0$ and vice-versa. The sign flipping in these sets of models suggests some joint parameter exists as a linear combination of the parameters I have estimated, which these models are trying to fit.

5. DYNAMIC MODELS

The previous section’s static models requires a strict factor exogeneity assumption outlined in equation (12) for all firms, including the own-firm effect of the production function. In this section, I use a dynamic model to relax this assumption. I also introduce a vector of temporal regressors: an autoregressor of own-firm GVA as a component of its TFP, and allowing for the past peer TFP components I analysed in the static model to influence current-period TFP in a lagged fashion. The structure of this specification now resembles a vector panel, which I will denote with $\text{VAR}(p,q)$ ²³.

$$z_{i,t} = \alpha_i + \gamma_t + \beta_0 + \sum_{k=1}^p \phi_k y_{i,t-k} + \sum_{k=0}^q \mathbf{z}_{t-k}^{\text{spill}} \boldsymbol{\theta}_k + \varepsilon_{i,t} \quad (32)$$

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \sum_{k=1}^p \phi_k y_{i,t-k} + \varepsilon_{i,t} \quad (33)$$

$$+ \sum_{k=0}^q \left[\delta_k \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_k - \delta_k \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_k - \delta_k \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t} \right]$$

Adding the autoregressor $y_{i,t}$ is also highly desirable. The panel setup with firm fixed effects already accounts for own-firm factors, and the presentation of R^2 including and excluding variation accounted for by the fixed effects throughout this paper demonstrates this. In the peer effects setting, autoregressive lags may also help control for reverse causality. If $y_{i,t-1}$ is causal for $z_{j,t}$, or feasibly $z_{j,t-1}$ and $\mathbb{E}[z_{j,t-1}, z_{j,t}] \neq 0$, then the inclusion of $y_{i,t-k}$ eliminates that bias. I offer no *a priori* view on the strength of ϕ_1 , the autoregressor coefficient. A significant estimate would imply heavy bias to the static models, validating the use in this dynamic setting.

There are also significant theoretical advantages to introducing the q -lags of the spatial spillover. This specification with both spatial and temporal lags accounts for the possibility that the spillover effect from firm j ’s production factor has a lagged impact on i , conceptualisable as the formation of clusters from firms seeing the benefits of colocation over some time after a productivity shock. Without including the q lags within $z_{i,t}$, if $\mathbf{z}_{t-k}^{\text{spill}}$ is correlated with $\mathbf{z}_t$, then estimating δ_0 in a static model when the true effect is dynamic exposes estimation to omitted variable bias (OVB) by attribution of the lagged to the current period’s regressor.

In practice for this dataset however, it is unlikely that more-than-one period spillover lag is relevant. This is due to this analysis using Fame’s annual frequency

²³ q -lags refers to those on the peer spillover effect, rather than those of a moving average component as is traditional of a VARMA(p,q) specification

observations. As a result, I show the structural equations in an AR(1,0) structure so that the primary advancement on the static models are purely the use of the weak exogeneity assumption, and first-order autoregressor. **(Additional tests)**

$$y_{i,t} = \alpha_i + \gamma_t + \beta_0 + \beta_1 k_{i,t} + \beta_2 l_{i,t} + \phi_1 y_{i,t-k} + \varepsilon_{i,t} \quad (34)$$

$$+ \delta_k \sum_{j \neq i} w_{ij} y_{j,t} + (\xi_k - \delta_k \beta_1) \cdot \sum_{j \neq i} w_{ij} k_{j,t} + (\zeta_k - \delta_k \beta_2) \cdot \sum_{j \neq i} w_{ij} l_{j,t}$$

5.1. DYNAMIC PANEL IDENTIFICATION

The dynamic panel specification uses a autocorrelation error orthogonality assumption to find valid identification (35). This replaces (12) and instead allows for some moderate level of cross-sectional error correlation (36).

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{i,t-1}] = 0, \quad \forall t, t-1 \quad (35)$$

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t}] \neq 0, \quad \exists i, j \quad (36)$$

$$\mathbb{E}[\varepsilon_t] = 0, \mathbb{E}[\varepsilon_t^2] = \sigma_t^2, \mathbb{E}[\varepsilon_t | \Omega_{t-1}] = 0, \quad \forall i \quad (\text{w.n.}) \quad (37)$$

No autocorrelation of errors may generally be seen as a more credible assumption, as in a dynamic, correctly-specified model with the correct autoregressive properties, time autocorrelation as component of the error term are modelled out. Conceptually, we could also argue that firms make efficient decisions, so if there is a shock observed in this period, firms ensure that in the next, any expected factor inputs that are realised by that period fully take advantage of the shock so its effect is then captured by the input regressors. Any remaining residual must have been expected and so mean zero (formally a Martingale sequence). Further econometric analyses could guarantee this assumption by plotting patterns of errors and check for autocorrelation, which could come from moving average or omitted latent variable structures to the specification missing here. This specification can be identified in reduced panel form, which requires first differences to eliminate fixed effects, allowing me to retain the use of firm fixed effects.

Estimation under these dynamic panel conditions are developed by Anderson and Hsiao (1981), with further techniques expanded in Arellano and Bond (1991) and Blundell and Bond (1998) which improve efficiency. The estimator given in Arellano and Bond (1991) is designed to allow for endogeneity, limiting only to weak exogeneity as I do here. Internal instruments of the endogenous regressors are found using their lagged first differences of the log variables. Identification is improved in the System GMM approach given in Blundell and Bond (1998) by also including the first logged, not differenced, lag of the endogenous regressors

which meets the usual instrument conditions, as an instrument. It estimate this section's results with System GMM. Valid instruments are therefore theoretically derived as $\Delta x_{i,\tau-j}$ for any j if $x_{i,\tau}$ is a valid instrument.

I commence finding these valid instruments by assuming the structural specification (8) exhibits no autocorrelation in its error term $\varepsilon_{i,t}$, though I now allow for moderate cross-sectional correlation across firms.

$$\mathbb{E}[x_{i,t}\varepsilon_{i,t}] \neq 0, \quad \exists i \quad (38)$$

$$x_{i,t} = \begin{pmatrix} k_{i,t} & l_{i,t} & \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix} \quad (39)$$

$$\Delta x_{i,t} = \begin{pmatrix} \Delta k_{i,t} & \Delta l_{i,t} & \Delta \sum_{j \neq i} w_{ij,t} z_{j,t} \end{pmatrix}$$

I firstly relax the strict exogeneity assumption of the standard factors to weak exogeneity. There are two approaches to this: capital and labour could be fully flexible inputs, meaning they are optimised as part of the decision set at time t , of which the TFP error shock is a component. Or, capital and labour can be assumed to be predetermined: capital faces some time-to-build constraint where each firm allocates investment in the previous period based on the limited information available then, Ω_{t-1} . A similar sticky decision pattern is assumed of labour, perhaps through job searching and hiring frictions. I assume a new optimal firm decision on capital based on information, denoted by Ω_t which includes $\varepsilon_{i,t}$ for its respective t . This approach allows the discovery of valid instruments which are exogenous to $\varepsilon_{i,t}$ and $\Delta \varepsilon_{i,t}$. I justify an exclusion restriction for $\Delta k_{i,t-j} \forall j > 1$, as it is predetermined, and $\Delta l_{i,t-k}, k > 2$, as it is flexible.

$$K_{i,t} = I_{i,t} + dK_{i,t-1} \quad (40)$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* | \Omega_{t-1}] \quad (41)$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* | \Omega_{t-1}] \quad (42)$$

$$K_{i,t}, L_{i,t} \in \Omega_{t-1} \quad (43)$$

$$\mathbb{E}[\varepsilon_{i,t} | \Omega_{t-1}] = 0$$

$$\mathbb{E}[\varepsilon_{i,t} | \begin{pmatrix} \mathbf{k}_t \\ \mathbf{l}_t \end{pmatrix}] = 0_{(2 \times 1)}, \quad \forall i \quad (44)$$

$$\Delta y_{i,t} = \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \delta_0 \sum_{j \neq i} w_{ij} \Delta z_{j,t} + \Delta \varepsilon_{i,t} \quad (45)$$

$$K_{i,t} = I_{i,t} + dK_{i,t-1} \quad (46)$$

$$I_{i,t} = \mathbb{E}[I_{i,t}^* \mid \Omega_{t-1}] \quad (47)$$

$$L_{i,t} = \mathbb{E}[L_{i,t}^* \mid \Omega_t] \quad (48)$$

$$\varepsilon_{i,t} \in \Omega_t$$

$$\implies K_{i,t} \propto \epsilon_{t-1}$$

$$\implies L_{i,t} \propto \epsilon_t$$

$$\Delta \varepsilon_{i,t} = \varepsilon_{i,t} - \varepsilon_{i,t-1}$$

$$\Delta k_{i,t} = k_{i,t} - k_{i,t-1} \propto \varepsilon_{i,t-1} \varepsilon_{i,t-2}$$

$$\implies \mathbb{E}[\Delta k_{i,t} \Delta \epsilon_t] \neq 0$$

$$\implies \mathbb{E}[\Delta k_{i,t-1} \Delta \epsilon_t] = 0, \mathbb{E}[k_{i,t-1} \varepsilon_t] = 0 \quad (49)$$

$$\Delta l_{i,t} = l_{i,t} - l_{i,t-1} \propto \varepsilon_{i,t}, \varepsilon_{i,t-1}$$

$$\implies \mathbb{E}[\Delta l_{i,t-2} \Delta \epsilon_t] = 0, \mathbb{E}[l_{i,t-2} \varepsilon_t] = 0 \quad (50)$$

Devising an internal instrument requires a similar evaluation of the components of $y_{j,t}$ as in the static problem. There is a dynamic corollary to the static derivation

of the endogenous term $y_{i,t}$.

$$\begin{aligned} \mathbf{y}_t &= \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + \delta_0 \mathbf{W} \mathbf{y}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t + \boldsymbol{\varepsilon}_t \\ &\quad + \sum_{m=1}^p (\phi_m \mathbb{I} + \delta_m \mathbf{W}) \mathbf{y}_{t-m} + \sum_{m=1}^p (\xi_m - \delta_m \beta_1) \mathbf{W} \mathbf{k}_{t-m} + \sum_{m=1}^p (\zeta_m - \delta_m \beta_2) \mathbf{W} \mathbf{l}_{t-m} \end{aligned} \quad (51)$$

$$\begin{aligned} (\mathbb{I} - \delta_0 \mathbf{W}) \mathbf{y}_t &= \boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t \\ &\quad + \sum_{m=1}^p (\phi_m \mathbb{I} + \delta_m \mathbf{W}) \mathbf{y}_{t-m} \\ &\quad + \sum_{m=1}^p \mathbf{W} \left[(\xi_m - \delta_m \beta_1) \mathbf{k}_{t-m} + (\zeta_m - \delta_m \beta_2) \mathbf{l}_{t-m} \right] + \boldsymbol{\varepsilon}_t \end{aligned} \quad (52)$$

$$\begin{aligned} \mathbf{y}_t &= \sum_{k=0}^{\infty} \delta_0^k \mathbf{W}^k \left(\boldsymbol{\alpha} + \gamma_t \boldsymbol{\iota} + \beta_0 \boldsymbol{\iota} + \beta_1 \mathbf{k}_t + \beta_2 \mathbf{l}_t + (\xi_0 - \delta_0 \beta_1) \mathbf{W} \mathbf{k}_t + (\zeta_0 - \delta_0 \beta_2) \mathbf{W} \mathbf{l}_t \right. \\ &\quad + \sum_{m=1}^p (\phi_m \mathbb{I} + \delta_m \mathbf{W}) \mathbf{y}_{t-m} \\ &\quad \left. + \sum_{m=1}^p \mathbf{W} \left[(\xi_m - \delta_m \beta_1) \mathbf{k}_{t-m} + (\zeta_m - \delta_m \beta_2) \mathbf{l}_{t-m} \right] + \boldsymbol{\varepsilon}_t \right) \end{aligned} \quad (53)$$

$$\begin{aligned} y_{j,t} &= \cdots + \underbrace{\sum_{k=0}^{\infty} \left[\delta_0^k \mathbf{W}^{k+1} \right]_{(j,)} \begin{pmatrix} \mathbf{k}_t & \mathbf{l}_t \end{pmatrix} \begin{pmatrix} \xi_0 \\ \zeta_0 \end{pmatrix}}_{\text{contemporaneous exogenous}_j} \\ &\quad + \underbrace{\sum_{m=1}^p \sum_{k=0}^{\infty} \left[\phi_m \delta_0^k \mathbf{W}^k + \delta_m \delta_0^k \mathbf{W}^{k+1} \right]_{(j,)} \mathbf{y}_{t-m}}_{\text{entangled historical inertia \& peer effects}_j} \\ &\quad + \underbrace{\sum_{m=1}^p \sum_{k=0}^{\infty} \left[\delta_0^k \mathbf{W}^{k+1} \right]_{(j,)} \begin{pmatrix} \mathbf{k}_{t-m} & \mathbf{l}_{t-m} \end{pmatrix} \begin{pmatrix} \xi_m - \delta_m \beta_1 \\ \zeta_m - \delta_m \beta_2 \end{pmatrix}}_{\text{historical exogenous network effects}_j} \\ &\quad + \underbrace{\sum_{k=0}^{\infty} \left[\delta_0^k \mathbf{W}^k \right]_{(j,)} \cdot \boldsymbol{\varepsilon}_t}_{\text{error transformation}_j} \end{aligned} \quad (54)$$

For system GMM, I need to show that the reduced-form first difference

regressors are exogenous. The first-order autoregressor, $\Delta \ln(y_{it})$ is endogenous.

$$\Delta y_{i,t} = \Delta \gamma_t + \beta_1 \Delta k_{i,t} + \beta_2 \Delta l_{i,t} + \sum_{k=1}^p \phi_k \Delta y_{i,t-k} + \delta_0 \sum_{j \neq i} w_{ij,t} \Delta z_{j,t} + \Delta \varepsilon_{i,t} \quad (55)$$

$$\Delta \varepsilon_{i,t} = \varepsilon_{i,t} - \varepsilon_{i,t-1} \quad (56)$$

$$\begin{aligned} \Delta y_{i,t-1} &= \dots + \Delta \varepsilon_{i,t-1} \\ &= \dots + \varepsilon_{i,t-1} - \varepsilon_{i,t-2} \\ &\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] \neq 0 \\ \Delta y_{i,t-2} &= \dots + \varepsilon_{i,t-2} - \varepsilon_{i,t-3} \\ &\implies \mathbb{E}[\Delta y_{i,t-1} \Delta \varepsilon_{i,t}] = 0 \end{aligned}$$

This is confirmed by the Anderson-Rubin (AR(1)) test in my results, which rejects the null hypothesis at the 1% level for all models, suggesting that first-order lags of log GVA are a weak instrument (Why? Shouldn't they fail exclusion restriction, not instrument correlation). In addition, I assume that capital, $k_{i,t}$ is pre-determined, as investment is chosen one period ahead, in line with standard Perpetual Inventory Method approaches²⁴ (Review predetermined material. Additionally, explain the logic of instrumenting for capital)

To overcome this endogeneity, I introduce deeper autoregressive lags as instrumental variables, as proposed by Anderson and Hsiao (1981). I use 3 autoregressive instrument lags starting from $t-2$: $\ln(y_{i,t-2})$, $\ln(y_{i,t-2})$, $\ln(y_{i,t-p-3})$ for log GVA, and 2 lags for capital $\ln(k_{i,t-p-1})$, $\ln(k_{i,t-p-2})$. An Anderson-Rubin (AR(2)) test shows that deep lags are a valid instrument for log GVA. The test applied to the second lag of $\ln(gva_1)$ fails to reject the null hypothesis of for any specification, so I assume it is a valid instrument.

Additionally, using multiple deep lags allows for overidentification: in this basic AR(1) specification, I have $k = 4$ regressors and $l = 7$ instruments. I could increase this. Fame is rich on the time dimension, with up to 19 lags at most for a firm, and a median of 6 lags, so I could safely provide more lags of regressors as instruments. Further lags is likely to have a decreasing marginal effect on explanatory power and additionally ((...))

I estimate the equation fulfilling the vector of moment conditions, (57) via

²⁴ Noting that this is ordinarily done on a quarterly basis, however, due to data limitations we adopt the same thing here even though Fame provides observations annually.

GMM.

$$\mathbb{E} \left[\Delta \varepsilon_{it} \cdot \begin{pmatrix} \Delta \ln(y_{i,t-2}) \\ \Delta \ln(y_{i,t-3}) \\ \Delta \ln(y_{i,t-4}) \\ \Delta \ln(k_{i,t-2}) \\ \Delta \ln(k_{i,t-3}) \\ \Delta \sum_j w_{ijt} z_{jt} \end{pmatrix} \right] = 0 \quad (57)$$

Finally, a Sargan-Hansen J-test is implemented to check if these instruments are over-identified, meaning the moment conditions cannot be fulfilled with 95% confidence. In all models, the J-test fails to reject the null hypothesis, meaning the model's moment conditions are jointly correct.

(!!!!) Following above reasoning, lags of $\Delta \sum_{j \neq i} w_{i,j,t} z_{j,t}$ of a higher order than 2 are exogenous to my reduced-form specification. However,

5.2. INSTRUMENTAL IDENTIFICATION UNDER ERROR AUTOCORRELATION

I appear to have strong identification given assumptions (35) and (36). The instruments above exhibit strong correlation under Anderson-Rubin tests and exclusion restriction has been derived. However, there exists a niche case of diagonally-correlated errors, namely despite (35) and (36), I ought to consider the following:

$$\mathbb{E}[\varepsilon_{i,t} \varepsilon_{j,t-k}] \neq 0, \quad \exists i, j, k \quad (58)$$

This is not as implausible as it may appear due to this analysis' peer effects setting. There may be some local omitted shock ν_t , that impacts peer firm A's TFP via ε_t and impacts target firm B's TFP after a lag via ε_{t-k} . This could be a price shock with random adjustment (or)

Additionally, relaxing (35) to allow for weak error autocorrelation between limited $\varepsilon_{i,t}$ and $\varepsilon_{i,t-k}$ for limited k proves a similar problem.

(Firstly, I test for error autocorrelation.)

TABLE VIII
Dynamic panel: baseline instrumental specification

	(1.) dyn1	(2.) dyn2
β_1	0.341 (0.123)	0.352 (0.145)
β_2	0.700 (0.249)	0.510 (0.237)
ϕ_1	-0.001 (0.001)	-0.001 (0.001)
δ_0	-0.076 (0.390)	0.195 (0.390)
ξ_0	-0.193 (0.239)	-0.043 (0.278)
ζ_0	0.511 (0.489)	0.493 (0.391)
J-test	X (0.834)	X (0.913)
AR(1) test	✓ (0.000)***	✓ (0.000)***
AR(2) test	X (0.129)	X (0.294)
Instr.	$\mathbf{y}_{t-2:3}, \mathbf{k}_{t-1:2}$ $\mathbf{l}_{t-2:3}, \mathbf{W}_{d1}\mathbf{y}_{t-2:3}$ $\mathbf{W}_{d1}\mathbf{k}_{t-1:2}, \mathbf{W}_{d1}\mathbf{l}_{t-2:3}$	$\mathbf{y}_{t-2:3}, \mathbf{k}_{t-2:3}$ $\mathbf{l}_{t-2:3}, \mathbf{W}_{d1}\mathbf{y}_{t-2:3}$ $\mathbf{W}_{d1}\mathbf{k}_{t-2:3}, \mathbf{W}_{d1}\mathbf{l}_{t-2:3}$
Obs.	717,300	717,300

Hansen J-test for over-identification: ‘X’ indicates insufficient evidence at the 95% level to reject the null-hypothesis that the model is identified.

Anderson-Rubin AR(1) test: ‘✓’ indicates the use of single-order lags is a weak instrument, as expected from a first-differencing SystemGMM approach.

Anderson-Rubin AR(2) test: ‘X’ indicates insufficient evidence to reject the use of second-order lags due to their being a weak instrument, as expected from the first-differencing with lags greater than 2nd-order to overcome contemporaneous endogeneity. ‘checkmark’ indicates some instruments used were weak.

6. EXTENSIONS

6.1. ALTERNATIVE DATA SOURCES FOR LOCATION DATA

6.2. SELECTION OF PARAMETERS

In section ??, I recorded quality of location data for each firm, but discussed limitations of applications of that data.

In section 3.5, I discussed arbitrary choices over network and distance restrictions, using *ttwa* boundaries and 5km max distances.

6.3. NETWORK FIXED EFFECTS

Firm and time fixed effects in a network setting may be considered insufficient, A local demand shocks, which can be thought of as a *correlated* shock $\mathbb{E}[\varepsilon_{i,t,g}\varepsilon_{i,t,h}] \neq 0$, where (g, h) are some close members of a network or group within the interactions matrix $\mathbf{W}$. This is a phenomenon specific to peer effects analysis and differs from the usual omitted variable bias (OVB) that impacts all firms discussed in standard econometric theory; here, specific identity of a group or network exposes members to a certain third-party common shock, for which a researcher should introduce controls. Manski (1993) labels these *correlated effects*. Econometrically, a *correlated* shock is some omitted variable that hits both error terms, such as a demand shock in the local area $v_{i,t} \rightarrow \varepsilon_{i,t}, \varepsilon_{j,t}$. This would simultaneously lift the recorded profit and GVA, of both a target and its peers. Another instance could be an exogenous change in the mutual supply chain of a firm and its peer, which is most likely to occur when i, j share an industry. A negative price cost shock, for instance, would lower the cost of inputs for both firms and be accounted for in the short run by increased GVA for the value of inputs indexed using macroeconomic data.

It may be considered an arbitrary choice what is a sufficiently ‘local’ region to capture these effects and model them as a variable correlated with regressors. Fortunately, econometric theory through Bramoullé et al. (2009), discusses the use of *local* network fixed effects which can be applied to existing $\mathbf{W}$ weighting matrices. Network-fixed effect reduced-form regressions are estimated by taking $(\mathbb{I} - \mathbf{W})$ transformations of all parameters: dependent variable, regressors, and instruments. In my analysis, this means the existing applied weighting matrices $\mathbf{W}_g, \mathbf{W}_d \dots$ for static network and group approaches respectively. A corollary for *global* network fixed effects is the use of a similar transformation $(\mathbb{I} - \mathbf{V})$ which

follows the assignment rule:

$$w_{ij} \in \mathbf{V} = \begin{cases} 0, & \text{if } w_{ij,l} \neq 0, \exists l < k \\ \frac{1}{s_g}, & \text{else if } (i, j) \in g \\ 0, & \text{otherwise} \end{cases}$$

This shares notable similarities with the group matrix $\mathbf{W}_g$: it is a ‘leave-on-in’, average of peers group transformation, demeaned. For application to the group model, $\mathbf{V}$ and $\mathbf{W}_g$ are near-identical, except for the use of s_g instead of s_{g-1} as the denominator for $\mathbf{V}$, and a target’s firm variable of interest is taken as part of the weighted average of its own peers. In extensions on section 3.5, it is n_g is calculated at the *ttwa* level to align with all firms within the mathematical ‘network’ of the set of distance-weighted firms.

As briefly laid out in section ??, identification holds through instruments $(\mathbb{I} - \mathbf{W})\mathbf{W}^2x$ on the endogenous $(\mathbb{I} - \mathbf{W})\mathbf{W}y$ regressor under certain conditions of $\mathbf{W}$. The formal test is linear independence of the $\mathbf{I}, \mathbf{W}, \mathbf{W}^2, \mathbf{W}^3$ and the same limits to computational feasibility apply to this higher-order ($n \times n$) set of matrices evaluation. Nevertheless, Bramoullé et al. (2009) suggests that more than $s_g \geq 3$ distinct group sizes assumed to be exogenous, or $s_d \geq 3$ network diameter should be sufficient conditions to the linear independence of the high-order matrices. The equivalent argument applies for *global* network fixed effects: they overcome the ‘Reflection Problem’ via the instrument $(\mathbb{I} - \mathbf{V})\mathbf{W}^2x$ and its higher-order equivalents.

$$\begin{aligned} (\mathbb{I} - \mathbf{W})\mathbf{y}_t &= \gamma_t(\mathbb{I} - \mathbf{W})\boldsymbol{\iota} + \beta_0 + \beta_1(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{k}_t + \beta_2(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{l}_t \\ &+ \beta_3(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{y}_t + \beta_4(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{k}_t + \beta_5(\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{l}_t + (\mathbb{I} - \mathbf{W})\boldsymbol{\varepsilon}_t \end{aligned} \quad (59)$$

$$0_{1 \times 2} = \mathbb{E}[\varepsilon_{i,t} \begin{pmatrix} (\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{k}_t & (\mathbb{I} - \mathbf{W})\mathbf{W}\mathbf{l}_t \end{pmatrix}]$$

$$\begin{aligned} (\mathbb{I} - \mathbf{V})\mathbf{y}_t &= \gamma_t(\mathbb{I} - \mathbf{V})\boldsymbol{\iota} + \beta_0 + \beta_1(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{k}_t + \beta_2(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{l}_t \\ &+ \beta_3(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{y}_t + \beta_4(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{k}_t + \beta_5(\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{l}_t + (\mathbb{I} - \mathbf{V})\boldsymbol{\varepsilon}_t \end{aligned} \quad (60)$$

$$0_{1 \times 2} = \mathbb{E}[\varepsilon_{i,t} \begin{pmatrix} (\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{k}_t & (\mathbb{I} - \mathbf{V})\mathbf{W}\mathbf{l}_t \end{pmatrix}]$$

This specification omits a firm fixed effect component α_i in equations (59) and (60), and accordingly has no need to first-difference the reduced form equation. This is due to the risk of applying of both firm and network fixed effects simultaneously to be multicollinear. I run the extension models (59) and (60) for

the static group and network weights $\mathbf{W} \in \mathbf{W}_{g1}, \mathbf{W}_{d1}$. Despite efforts with the above, I find that all specifications failed the Sargan-Hansen J test, which implies over-identifying restrictions on instruments such that the GMM-CUE estimator could not converge to the required zero moment conditions. Estimates also suffer from multicollinearity anyway, giving parameter coefficients which fall far out of the expected range of $|\delta_0|, |\xi_0|, |\zeta_0| < 1$. Unstable estimates such as these should not be considered consistent, so I do not report them here.

Further analysis could investigate this issue more thoroughly and successfully implement these network fixed effects as the key to stripping out local demand effects or supply shocks.

6.4. DRIVERS OF THE STRENGTH OF SPILLOVER

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A. FAME DATA COMPUTATIONAL METHODS

The following section provides detailed guidance for the data manipulation undertaken in this analysis. I emphasise reproducibility due to the discussion on section 1 ‘Introduction’: this analysis offers a substantive baseline with advanced data science techniques and econometric methods, on top of which I leave it for future analysis to carry out the more sophisticated extensions of section 6. All scripts are publicly available online²⁵, written in Python and Jupyter notebooks.

A.1. DATA CLEANING PROCEDURE

DuckDB operates locally, meaning minimal lag from issuance to execution time provided the operating machine is sufficiently high-compute, and stores data in column vectors, which is a significant advantage to manipulating whole variables. Any SQL commands issued from the Python are compiled down to C, making execution for column transformation operations extremely fast, even for joining tables of several 100s of millions of rows²⁶, as is calculated in this analysis for target-peer spatial distances. I wrote database interaction commands using *Ibis*, an Apache-2.0 licensed Object-Relational Mapping (ORM) framework which allows SQL commands to be written natively in Python²⁷, minimising the likelihood of errors when issuing queries loops or changing variable names. This architecture choice is extremely well-suited for mass data econometric analysis and critical to solving the compute limitations of iterative large- n dataset manipulations required to calculate the valid instruments identified in section 4.1.

Raw data are distributed across a series of Excel files on balance sheet indicators, including time-invariant characteristics for each firm, bucketed into *a1_ID* and *a5_misc* files, and annual data published on a calendar-year basis, bucketed into *a2_key_finance*, *a3_assets*, and *a4_profits* tables.

I traversed the file structure to build a dictionary to the filepaths organised in their subfolders, which are named by the 2-digit industry SIC division, and batch-imported data belonging to the firms in that industry. Some industries have prodigious registered firms, such as SIC division 47 (Retail Trade, 1090 files) and 82 (Office administrative, office support and other business support activities, 1065 files)²⁸. I imposed a batch limit of 200 for stable memory handling, splitting up these large industries, and used Python’s asynchronous *concurrent.futures* package to execute the ingestion of each batch, exploiting multithreaded processing. In total, 12,800 files were processed across the 88 available SIC divisions. Time-

²⁵ <https://github.com/alloysiuslip/ucl-diss-ECON0079>

²⁶ Explanatory numbers given in this section are rounded to 3 significant figures for ease

²⁷ The Ibis Project Authors (2026)

²⁸ Office for National Statistics (2026d)

invariant firm characteristics were saved to a table named *fixed*, annual firm data to a table named *yearly* within *DuckDB*.

All firm data are tagged with a *registered_number* property corresponding to a Companies House ID. I deduplicated 896,000 *fixed* firm entries using a *registered_number* match, of which the vast majority were exact duplicate entries and the remainder had some mismatch of 1 or more firm characteristics, such as the listed company name, global operating owner, and in rare cases, location. Duplicate entries are an undesirable feature of the Fame dataset, worsened by Fame’s construction from snapshots of reported firm data across timepoints, where resolution of conflicting *fixed* data may be imperfect. Fame also contains data impute errors, meaning variations in names referring to the same entity. When duplicates occurred, I effectively randomly chose an entry to keep. A more robust analysis could devise further data science methods, such a deployment of an acceptable Levenshtein distance to resolve differing name entries, or separately treating firms with multiple listed addresses. Further data cleaning procedures are discussed in Irastorza-Fadrique et al. (2026), notably these include avoiding double-counting of firms through both their parent company and subsidiaries, truncation of data to at earliest 2010 due to changes in reporting standards, and reclassification of firm SIC codes using NLP on firm business activity descriptions.

As the main analysis here only uses location data as the *fixed* characteristic of analysis, due to its time constraints, and due to the low number of duplicates and other cleaning errors listed above relative to the size of the dataset, I treat those errors as random effects and deem the impact of the random selection strategy minimal. I similarly removed duplicate firm-year entries in the *yearly* data, and merged the two tables based on ensuring every *fixed* firm identity had a corresponding *yearly* set of rows, and vice versa. For efficiency, I adopted a ‘max-merge’ column strategy, which is effective for resolving entries which are zero and non-zero in conflicting duplicates. This however means that any two observations with conflicting non-zero values keep the large value. If it is random errors causing Fame to have these conflicting, duplicate values, this strategy has the potential to bias all financial data entries upwards, however this bias is unlikely to be correlated with any of this investigation’s variables of interest. I further removed a small minority (<200) firm entries not based in the UK, which can be identified through their *registered_number* format. In total, deduplication reduced the number of firms from 9,280,000 to 8,220,000, and the number of panel observations from 44,900,000 to 42,900,000.

I restrict the sample to observations with more than 10 employees, and drop any observation from the *yearly* data missing *renumeration_employees* and *ebitda*

properties which are required to calculate firm GVA. This follows Irastorza-Fadrique et al. (2026): low-employee firms, additionally more likely to be shell companies, are likely to mechanically yield an inaccurately high TFP-residual calculation. Doing this at the observation rather than firm, has the risk of inducing bias to the variables measured as there may remain firms in the sample straddling the employees threshold which endogenously have their observation years of low-employees excluded. First-differencing in panel regressions with this data provides some mitigation to this issue by providing a buffer of one year either side to excluded observations. Nevertheless, this analysis adopts the observation-level threshold strategy. This set of filters reduced the number of valid firms in this analysis to 152,000 with 1,130,000 panel observations.

Figure VI shows the distribution of active firms across years, which follows

FIGURE VI
Panel observation availability by year

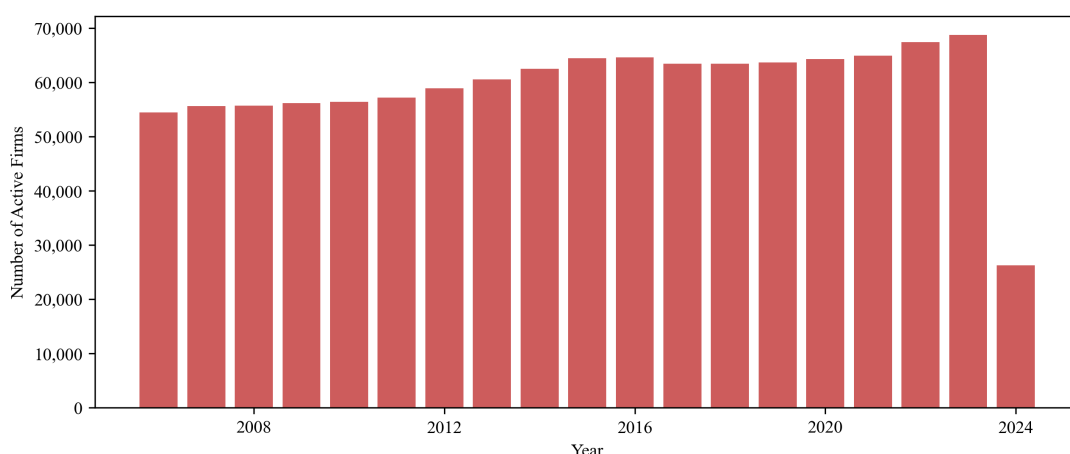
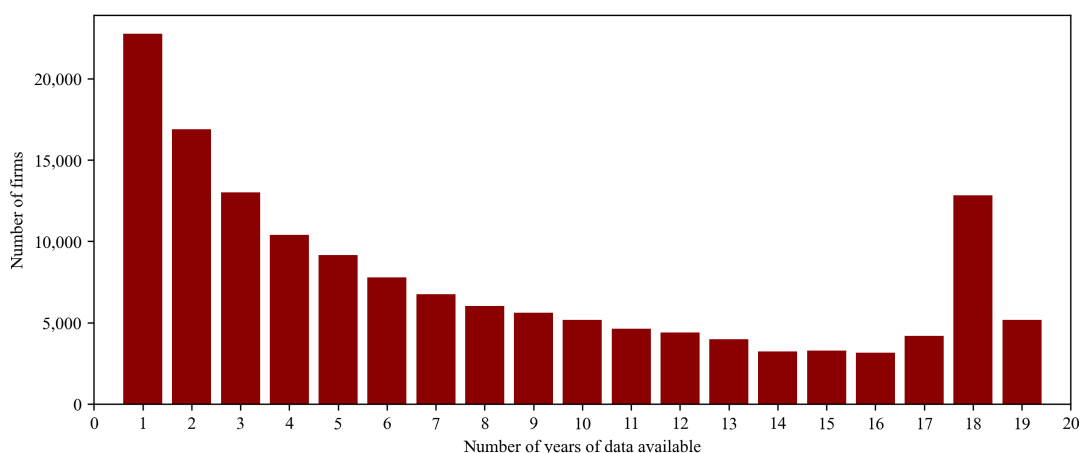


FIGURE VII
Distribution of years of data available per firm



Fame panel sample downloaded from calendar years 2006 to 2024 inclusive. Firms with zero years of data available filtered out. Sample downloaded February 2025, resulting in incomplete reporting for 2024.

expected trends from macroeconomic business population surveys²⁹. We can note an observable dip after the 2016 Brexit referendum and during the COVID-19 pandemic. We can also observe that 2024 is an incomplete sample set as not all firms had submitted their financial accounts for that year, but this is treated as a random effect and no further firms are filtering as a result of this.

The lifespan of a firm, and the corresponding number of *yearly* observations within the dataset follows a geometric distribution as given in Figure VII. With each passing year, a firm has a binary survival-remain decision, with an increasing discrete probability of survival for older firms.

A.2. VECTOR TRANSFORMATIONS PROCEDURE

Computation of a matrix product, including taking higher indices, of the interactions between 1,520,000 firms is not computationally difficult, especially with sophisticated algorithms to handle $\mathbf{W}$'s block diagonal structure. However, storing the calculation in temporary memory is beyond the capacity of any standard machine available as a research tool.

Instead, I use a computational approach that avoids calculating the $\mathbf{W}^2, \mathbf{W}^3 \dots$ matrix separately, but rather iteratively applies the last pre-multiplied matrix ($n \times n$) to the aggregate variable ($n \times 1$). This allows repeated use of the transformation matrices $\mathbf{W}_{g1}, \mathbf{W}_{d1}, \dots$ without having to compute a new transformation matrix $\mathbf{W}^2 \dots$. Additionally, the 'native' matrices $\mathbf{W}$ are easier to apply as transformations as their weight values can be computed on-demand using the group-network structure procedures, whereas each entry of an entirely new matrix $\mathbf{W}^2$ would have to be stored in its own table.

Using memory-efficient *DuckDB*, I designed the following procedure:

1. Establish a set of transforms that could be applied many-to-many to a set of vectors.

Ex 1: {variables : y, k, l , transforms : $g1, g2, g3$ }

Ex 2: {variables : $wg1_k, wg1_l$, transforms : $g1$ }

2. Parse this many-to-many relation to a one-to-one relation, that maps onto the new column vector post-transform. This requires a function with a consistent set of rules to apply to a transformation-vector combination. Map each entry in the one-to-one relation to a 'depth' level, corresponding to number of new vectors that need to be created first before the right vector exists to apply the latest transformation.

²⁹ Department for Business and Trade (2025)

```

Depth 1 (15): [ 'y -> wg1_y', 'y -> wg2_y', 'y -> wg3_y', ... ]
Depth 2 (24): [ 'wg1_k -> w2g1_k', 'wg1_k -> wg2wg1_k', ... ]
Depth 3 (10): [ 'w2g1_k -> w3g1_k', 'w2g1_l -> w3g1_l', ... ]
Depth 4 (10): [ 'w3g1_k -> w4g1_k', 'w3g1_l -> w4g1_l', ... ]
Depth 5 (4): [ 'w4g1_k -> (i-wg1)w4g1_k', ... ]

```

3. Execute the transformations at each depth level in parallel, calculating the table of vectors after each depth as an input to the new depth set of transformations.

Group W_g : sum the target vector over i firms in the panel database, adopting a ‘leave-one-out’ approach as necessary by subtracting the current row. Row-normalise by counting the summed rows and taking it as the quotient denominator.

Group W_d : Pre-calculate the distance transformation measures d_1, d_2, d_3 on the distance table and join the panel table to it twice: once for the target $firm_i$ and once for the peer $firm_j$. Take the specified weight column for the transformation (ex: d_1) and multiply it with the specified variable vector. Row-normalise by summing the transformed-distance (weight) column and using as a denominator.

A.3. W TRANSFORMATIONS SUMMARY STATISTICS

TABLE IX
Variable names from transforming $\mathbf{y}_t$ and $\mathbf{x}_t$ regressors via weighting matrix $\mathbf{W}$

		endogenous	exogenous	
	Matrix product	gva	capital	labour
<i>Groups</i>				
regressors	$\mathbf{W}_{g1}$	wg1_y	wg1_k	wg1_l
	$\mathbf{W}_{g2}$	wg2_y	wg2_k	wg2_l
	$\mathbf{W}_{g3}$	wg3_y	wg3_k	wg3_l
instruments	$\mathbf{W}_{g1}^2$		w2g1_k	w2g1_l
	$\mathbf{W}_{g1}\mathbf{W}_{g2}$		wg1wg2_k	wg1wg2_l
	$\mathbf{W}_{g2}\mathbf{W}_{g3}$		wg2wg3_k	wg2wg3_l
	$\mathbf{W}_{g3}\mathbf{W}_{g1}$		wg3wg1_k	wg3wg1_l
	$\mathbf{W}_{g2}\mathbf{W}_{g1}$		wg2wg1_k	wg2wg1_l
	$\mathbf{W}_{g3}\mathbf{W}_{g2}$		wg3wg2_k	wg3wg2_l
	$\mathbf{W}_{g3}\mathbf{W}_{g1}$		wg3wg1_k	wg3wg1_l
local network FE	$(\mathbb{I} - \mathbf{W}_{g1})\mathbf{W}_{g1}^2$		(i-wg1)w2g1_k	(i-wg1)w2g1_l
global network FE	$(\mathbb{I} - \mathbf{V}_{g1})\mathbf{W}_{g1}^2$		(i-vg1)w2g1_k	(i-vg1)w2g1_l
<i>Network</i>				
regressors	$\mathbf{W}_{d1}$	wd1_y	wd1_k	wd1_l
	$\mathbf{W}_{d2}$	wd2_y	wd2_k	wd2_l
	$\mathbf{W}_{d3}$	wd3_y	wd3_k	wd3_l
instruments	$\mathbf{W}_{d1}^2$		w2d1_k	w2d1_l
	$\mathbf{W}_{d2}^2$		w2d2_k	w2d2_l
	$\mathbf{W}_{d3}^2$		w2d3_k	w2d3_l
	$\mathbf{W}_{d1}^3$		w3d1_k	w3d1_l
	$\mathbf{W}_{d1}^4$		w3d2_k	w3d2_l
local network FE	$(\mathbb{I} - \mathbf{W}_{d1})\mathbf{W}_{d1}^4$		w3d3_k	w3d3_l
global network FE	$(\mathbb{I} - \mathbf{V}_{d1})\mathbf{W}_{d1}^4$		w4d1_k	w4d1_l

Total: 51 new variables to be created, for a total of 53 columns in a linked table.

B. STATIC GROUPS: ADDITIONAL RESULTS

TABLE X
First-stage 2SLS (panel OLS) results

	(1.) lag2	(2.) lag3	(3.) lag4
const	4.152 (0.046)	1.896 (0.030)	2.330 (0.028)
$\mathbf{W}_{g1}^2 \mathbf{k}$	0.263 (0.005)	0.048 (0.005)	-0.660 (0.026)
$\mathbf{W}_{g1}^2 \mathbf{l}$	0.433 (0.007)	0.088 (0.005)	-1.132 (0.029)
$\mathbf{W}_{g1}^3 \mathbf{k}$		0.324 (0.006)	0.327 (0.006)
$\mathbf{W}_{g1}^3 \mathbf{l}$		0.603 (0.006)	0.614 (0.006)
$\mathbf{W}_{g1}^4 \mathbf{k}$			0.679 (0.025)
$\mathbf{W}_{g1}^4 \mathbf{l}$			1.169 (0.028)
Obs.	726,283	726,283	726,283
R ² -adj (excl. F.E.)	0.2178	0.5632	0.5887
R ² -adj (Overall)	0.4646	0.7812	0.8104

C. STATIC DISTANCE-DECAY: ADDITIONAL RESULTS

TABLE XI
Static panel IV, 3 orthogonal groups

	(1.) exog-2	(2.) iv-2	(3.) exog-3	(4.) iv-3
β_1	0.261 (0.005)***	0.233 (0.008)***	0.261 (0.005)***	0.236 (0.011)***
β_2	0.650 (0.005)***	0.555 (0.008)***	0.649 (0.005)***	0.580 (0.020)***
δ_1	0.198 (0.004)***	-0.337 (0.191)*	0.198 (0.004)***	-0.715 (0.596)
δ_2	0.020 (0.006)***	2.812 (1.038)***	0.019 (0.006)***	4.312 (3.073)
ξ_1	-0.006 (0.002)**	0.025 (0.015)*	-0.006 (0.002)**	0.047 (0.043)
ζ_1	0.011 (0.003)***	0.037 (0.015)**	0.011 (0.003)***	0.048 (0.030)
ξ_2	-0.007 (0.004)*	-0.393 (0.143)***	-0.007 (0.004)*	-0.477 (0.375)
ζ_2	0.011 (0.006)*	-0.203 (0.088)**	0.011 (0.006)*	-0.331 (0.179)*
δ_3			0.048 (0.017)***	-40.411 (25.266)
ξ_3			-0.009 (0.010)	5.323 (3.293)
ζ_3			0.019 (0.016)	2.586 (1.100)**
J-test		✓ (0.017)**		X (0.559)
Instr.		$\mathbf{W}_{g1}^2 \mathbf{k}, \mathbf{W}_{g1}^2 \mathbf{l}$ $\mathbf{W}_{g2}^2 \mathbf{k}, \mathbf{W}_{g2}^2 \mathbf{l}$		$\mathbf{W}_{g1}^2 \mathbf{k}, \mathbf{W}_{g1}^2 \mathbf{l}$ $\mathbf{W}_{g2}^2 \mathbf{k}, \mathbf{W}_{g2}^2 \mathbf{l}$ $\mathbf{W}_{g3}^2 \mathbf{k}, \mathbf{W}_{g3}^2 \mathbf{l}$
Obs.	719,403	581,797	718,743	581,185
R ² (excl. F.E.)	0.3543	-0.4604	0.3543	-11.5040
R ² (Overall)	0.7967	0.0000	0.7970	0.0000

TABLE XII
First-stage 2SLS (panel OLS) results

	(1.) lag2	(2.) lag3	(3.) lag4	(4.) lag2-noz	(5.) lag3-noz	(6.) lag4-noz
const	4.099 (0.045)	1.691 (0.025)	1.805 (0.024)	3.913 (0.052)	1.568 (0.033)	1.685 (0.034)
$\mathbf{W}_{d1}^2 \mathbf{k}$	0.277 (0.005)	0.024 (0.004)	0.355 (0.013)	0.312 (0.007)	-0.016 (0.008)	0.185 (0.014)
$\mathbf{W}_{d1}^2 \mathbf{l}$	0.397 (0.007)	0.025 (0.005)	0.144 (0.018)	0.361 (0.009)	-0.064 (0.010)	0.046 (0.018)
$\mathbf{W}_{d1}^3 \mathbf{k}$		0.373 (0.005)	0.392 (0.005)		0.440 (0.011)	0.470 (0.012)
$\mathbf{W}_{d1}^3 \mathbf{l}$		0.640 (0.006)	0.646 (0.006)		0.707 (0.013)	0.722 (0.014)
$\mathbf{W}_{d1}^4 \mathbf{k}$			-0.361 (0.014)			-0.239 (0.018)
$\mathbf{W}_{d1}^4 \mathbf{l}$			-0.124 (0.020)			-0.132 (0.025)
Obs.	1,031,359	1,031,359	1,031,359	1,025,318	1,025,318	1,025,318
R ² -adj (excl. F.E.)	0.2190	0.5901	0.5964	0.1835	0.5280	0.5329
R ² -adj (Overall)	0.4532	0.7551	0.7639	0.4105	0.7239	0.7322

TABLE XIII
Static panel IV using continuous distance-decay $\mathbf{W}_d$, zero
distances excluded

	(6.) dd1-noz	(7.) dd2-noz	(8.) dd3-noz	(9.) dd1-deep3-noz
β_1	0.273 (0.007)***	0.273 (0.007)***	0.270 (0.007)***	0.273 (0.007)***
β_2	0.509 (0.006)***	0.510 (0.006)***	0.516 (0.007)***	0.509 (0.006)***
δ_0	0.776 (0.487)	0.463 (0.203)**	-4.083 (2.512)	1.058 (0.363)***
ξ_0	-0.080 (0.051)	-0.040 (0.018)**	0.450 (0.281)	-0.109 (0.038)***
ζ_0	-0.085 (0.060)	-0.054 (0.026)**	0.480 (0.285)*	-0.119 (0.045)***
J-test	X (0.381)	X (0.609)	X (0.074)*	X (0.672)
Instr.	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$	$\mathbf{W}_{d2}^2 \mathbf{k}, \mathbf{W}_{d2}^2 \mathbf{l}$	$\mathbf{W}_{d3}^2 \mathbf{k}, \mathbf{W}_{d3}^2 \mathbf{l}$	$\mathbf{W}_{d1}^2 \mathbf{k}, \mathbf{W}_{d1}^2 \mathbf{l}$ $\mathbf{W}_{d1}^3 \mathbf{k}, \mathbf{W}_{d1}^3 \mathbf{l}$
<i>i</i> F.E.	✓	✓	✓	✓
<i>t</i> F.E.	✓	✓	✓	✓
Obs.	848,718	848,718	848,718	848,718
R ²	0.0712	0.0754	-1.5298	0.0269

D. DYNAMIC PANEL: ALTERNATIVE SPECIFICATIONS